Adaptive Hypergraph Regularized Multilayer Sparse Tensor Factorization for Hyperspectral Unmixing

Pan Zheng, Hongjun Su, *Senior Member, IEEE*, Hongliang Lu, and Qian Du, *Fellow, IEEE*

***Abstract*— Hyperspectral unmixing with tensor models has received great attention in recent years. A tensor-based decomposition method can effectively represent the structural feature of hyperspectral images; however, the obtained results may be physically uninterpretable. To overcome this limitation, a novel adaptive hypergraph regularized multilayer sparse tensor factorization (AHGMLSTF) algorithm is proposed. First, a modified hypergraph is incorporated into tensor factorization, and the modified hypergraph uses spectral angle distance (SAD) instead of Euclidean distance to construct hyperedges to better represent the joint spatial and spectral information. Then, the hypergraph is constructed adaptively by hyperedges of k neighborhoods. Second, the concept of multilayer decomposition is introduced to explore the hierarchical features of hyperspectral images, and a sparse constraint is imposed on each layer to make the unmixing results more consistent with the physical mechanism of mixed spectral pixels. With these constraints, the proposed method established a spectral–spatial joint tensor decomposition model that represents not only the local neighborhood similarity but also the heterogeneity of adjacent edges. Experiments on simulated data and real hyperspectral data demonstrate the effectiveness of the proposed method.**

***Index Terms*— Hypergraph, hyperspectral image, multilayer, sparse unmixing, tensor factorization.**

I. Introduction

HYPERSPECTRAL imagery combines the spectral information reflecting the material radiation with the spatial information of geomorphology. It has the characteristics of multiple bands, narrow spectral range, and continuous bands, greatly enhancing the ability of remote sensing images to detect material property information [1]. However, due to spatial resolution limitations, the complexity of land cover, and the mixing effect of the imaging sensor itself, some pixels contain multiple materials, and such pixels are called mixed pixels. A large number of mixed pixels make the development and application of hyperspectral image analysis more challenging [2]. Therefore, extracting the spectral features from mixed pixels and calculating the proportion of each spectral feature become the key preprocessing technique for hyperspectral images, which is called hyperspectral image unmixing. The extraction of spectral features contained in a mixed pixel is called endmember extraction, and calculating the proportion of each endmember is called abundance estimation [3], [4], [5]. The linear spectral mixture model (LSMM) has been widely used because of its simplicity, high efficiency, and good description of the actual spectral mixing process. Many effective unmixing algorithms are developed based on the LSMM [6].

In general, linear unmixing can be achieved by geometric methods, statistical methods, or sparse methods. Geometric methods treat the vertices of a simplex as endmembers, and it is mainly used to extract endmembers. N-finder (N-FINDR) [7] and vertex component analysis (VCA) [8] are the most popular geometric methods. N-FINDR automatically obtains endmembers in a hyperspectral image by searching for a simplex with the maximum volume. VCA extracts endmembers one by one by iteratively searching orthogonal vectors and calculating the projection distance of the image matrix on orthogonal vectors. Although these methods are simple, they assume the existence of pure endmembers, which may not be true [9]. After endmember extraction, a least-squares algorithm is also needed to obtain the abundance results, and the estimated endmembers directly affect the accuracy of unmixing [10].

Sparse unmixing via variable splitting and augmented Lagrangian (SUnSAL) has attracted much attention because of its excellent performance [11]. Sparse unmixing can directly estimate the abundance maps by using a spectral library without endmember extraction. Since a spectral library consists of a large number of spectral features from which a few spectral signatures are selected to approximate the mixed pixels, which makes abundances only contain a small number of nonzero elements, the abundance matrix has the sparse property. Many algorithms are derived from the SUnSAL algorithm. To improve the unmixing accuracy, current algorithms are improved mainly from two aspects: sparse prior of abundance and spatial context information. For the sparse prior information of abundance, the expression ability of sparse characteristics from l_1-norm and $l_{0.5}$-norm to l_p-norm is gradually strengthened. Considering that the spectral signature of mixed pixels in homogeneous regions has high similarity and the corresponding abundances are highly correlated, this prior information can be expressed by imposing an $l_{2,1}$-norm constraint on the abundance matrices, and this algorithm is called constrained sparse unmixing via variable splitting and augmented Lagrangian (CLSUnSAL) [12]. To accommodate the influence of spatial information on sparsity, two weight matrices are introduced to improve the row sparsity

Manuscript received 11 May 2022; revised 13 September 2022 and 31 December 2022; accepted 9 January 2023. Date of publication 31 January 2023; date of current version 9 February 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42122008 and Grant 41871220. *(Corresponding author: Hongjun Su.)*

Pan Zheng, Hongjun Su, and Hongliang Lu are with the School of Earth Sciences and Engineering, Hohai University, Nanjing 211100, China (e-mail: hjsu@hhu.edu.cn).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA.

Digital Object Identifier 10.1109/TGRS.2023.3241115

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

of abundance and the sparsity of abundance vector of each endmember [13]. The disadvantage of the abundance sparse prior algorithms is that they ignore the spatial structure in the hyperspectral data, as the norm is a measure of distance, and the similarity between structures cannot be measured. Another improvement for sparse unmixing is the use of spatial context information. SUnSAL-TV introduces the total variation as prior information into sparse unmixing, which strengthens the spatial uniformity between adjacent pixels [14]. SUnSAL-TV may lead to over-smoothing and blurred boundaries. Considering the global spatial information of the whole abundance map, central pixels are represented by the mean of similarly structured pixels in spectral space; this method effectively maintains the spatial correlation of hyperspectral images. Subsequently, the weighted algorithm and the adaptive mean version were proposed to improve the algorithm [15], [16]. The abundance matrix can be more accurately estimated using local and nonlocal information in algorithms based on spatial contextual information [17], [18], [19], [20], [21]. However, spatial information is difficult to model its variation. These improvements for sparse unmixing can yield better unmixing results, but the problem of poor migration of the spectral library still exists due to the difference between the library and the actual endmembers in a real image.

Nonnegative matrix factorization (NMF) is the most popular algorithm in statistical methods. NMF for hyperspectral unmixing directly decomposes an image matrix into the product of an endmember matrix and an abundance matrix. According to the physical meaning, these two matrices need to meet nonnegative attributes. In practical applications, the number of endmembers is much smaller than the number of bands, and this makes the solution of NMF an infinite problem. Therefore, NMF-based unmixing is mainly improved in three aspects by adding a priori information to achieve a unique solution: the prior information of endmember and abundance, the structural information, and the improvement combined with deep learning. The local smoothness of the endmember spectrum and the sparsity of the abundance can be used as prior information, and many algorithms have been developed accordingly [22], [23], [24]. The disadvantage of NMF algorithms with prior information is that the images cannot be accurately unmixed when the mixture is extremely similar to the endmember spectrum. The manifold structure is embedded into NMF unmixing model to maintain the local structure of the image and capture the similarity in the local area for unmixing [25], [26], and this idea makes use of the structural information of the image. Based on this algorithm, the dual-graph NMF is adopted to improve the unmixing accuracy by combining the internal manifold in spectral space with the consistency constraint of spatial information [27]. The disadvantage of these graph-based algorithms is that they can only represent the relationship between two pixels, and then, the hypergraph model is used to represent the relationship between spatial similarity and spectral similarity more accurately [28]. A variety of NMF unmixing algorithms based on the structure information are proposed [28], [29], [30], [31], [32]. The improved algorithms based on NMF relax all the constraints. However, these methods do not make full use of structural information when stretching a 3-D hyperspectral image into a matrix.

The third-order tensor preserves spatial information without distortion, which is more suitable for hyperspectral data. Therefore, tensor decomposition models have received widespread concern in hyperspectral data processing, including hyperspectral image reconstruction, hyperspectral image denoising, hyperspectral unmixing, and hyperspectral fusion [33], [34], [35], [36]. Canonical Polyadic (CP) decomposition, tucker decomposition, block term decomposition (BTD), and tensor train (TT) decomposition are introduced into hyperspectral processing. Tucker decomposition method decomposes a third-order tensor into three-factor matrices and a core tensor. Renard et al. [37] proposed a low-rank tensor approximation (LRTA) method, which uses tucker decomposition to obtain a low-rank approximation of hyperspectral images for denoising. Huck and Guillaume [38] first introduced tucker decomposition into hyperspectral unmixing and proposed a method to estimate three required data tucker ranks. CP decomposition decomposes a third-order tensor into a series of rank-one component tensors. By sorting each component after CP decomposition and selecting the first few components containing the main information for reconstruction, the hyperspectral image can be denoised. Zhang et al. [39] first introduced CP decomposition into hyperspectral unmixing. To make CP decomposition more robust and accurate, some prior knowledge can be added to the decomposed factors, for example, smooth constraints, orthogonal constraints, nonnegative constraints, and sparse constraints. However, the physical meanings of the results obtained by tucker decomposition and CP decomposition are not clear. Veganzones et al. [40] proposed the BTD method, which decomposes an N-order tensor into R component tensors. The BTD method is widely used in hyperspectral unmixing because it fits the physical meaning of the spectral mixing model. Different types of prior knowledge are incorporated into the BTD model, such as sparsity constraint, nonnegative constraint, and local spatial similarity. Recently, additional BT decomposition models have been proposed; Qian et al. [41] combined CP decomposition with BTD and proposed matrix-vector nonnegative tensor factorization (MV-NTF), which established the physical connection with linear mixing model. TV regularization term and sparse and low-rank constraints are applied to the MV-NTF model [42], [43]. BTD and MV-NTF models have achieved good results in the decomposing process. However, it is still challenging to effectively establish the spectral–spatial joint relation and accurately represent the data structure information [44], [45]. The idea of deep learning as multilayer decomposition is also introduced into tensor factorization [46]. Multilayer representation can encode the structured sparsity of the tensor, and some research has been carried out in tensor completion in recent years; Xue et al. [47] adopted the CP decomposition model, using the number of tensor components, the smoothness of the factor matrix, and low-rank matrix decomposition to construct three layers of sparsity for tensor completion. Subsequently, Xue et al. [48] improved tucker decomposition

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

and proposed a parametric tensor sparsity measure model to describe the hierarchical structure for tensor completion. These multilayer tensor decomposition models can be well-applied for tensor completion, but lack physical meaning for hyperspectral unmixing. At present, there are few multilayer tensor decomposition algorithms for hyperspectral unmixing.

To summarize, the spectral library for sparse unmixing has poor migration, and the norm for sparseness is a distance measure that cannot accurately measure the structure of the hyperspectral data. The NMF unmixing algorithms can directly decompose the matrix of the hyperspectral image, and the local extreme solutions can be avoided by adding prior information, but it is still difficult to accurately represent the structural information. Multilayer tensor decomposition models can represent the image hierarchy well, but it lacks physical meaning for hyperspectral unmixing. Based on these problems, how to incorporate the hyperspectral image structural information and establish a mathematical model of the joint spectral–spatial relationship for spectral unmixing is still an open problem.

Thus, in this article, a new tensor factorization model is proposed, which considers joint spectral–spatial information and hierarchical features simultaneously. To maintain the structural characteristics, the hyperspectral images are represented by a three-order tensor, and a matrix-vector tensor factorization algorithm is used for unmixing, which accords with the physical mechanism of mixed pixels. To represent the homogeneity of the local area and the heterogeneity of adjacent regions simultaneously, the hypergraph construction method is modified, in which the spectral angle distance (SAD) is used to replace the Euclidean distance, and the hypergraph is adaptively constructed with a set of hyperedges of k neighborhoods. Accordingly, the adaptive hypergraph regularized nonnegative tensor factorization (AHGNTF) algorithm for hyperspectral unmixing is proposed. To avoid falling into local minima, the concept of multilayer decomposition is introduced, the hierarchical character of the hyperspectral image is used, each decomposed layer contains less endmember and abundance information, and the sparsity of endmember and abundance matrices is considered in the decomposition process, thus resulting in the adaptive hypergraph regularized multilayer sparse tensor factorization (AHGMLSTF) algorithm. The contributions of this article are summarized as follows.

1) In the AHGNTF, the Euclidean distance is replaced by the SAD in the hypergraph construction process, which can not only utilize the local spatial information, but also effectively distinguish the edge information. Then, multiple sets of different k neighbor hyperedges are used to adaptively represent the local spatial–spectral information.
2) In the AHGMLSTF, the concept of multilayer NMF is introduced into tensor factorization to obtain hierarchical information in hyperspectral images. The modified hypergraph is constructed to capture the structure information of spatial and spectral domains in each layer. The sparsity of both endmember and abundance matrices is integrated in each layer.
3) In the process of model optimization, the objective function with hypergraph regularization and sparse constraint is solved by the alternating multiplier method (ADMM), which is beneficial to parallel processing. The multilayer decomposition is used to build a deep structure, and the error iteration is minimized to avoid local minima.

The rest of this article is arranged as follows. Section II describes the related work. Section III presents the proposed AHGNTF and AHGMLSTF algorithms. Experiments using simulated and real data are carried out to evaluate the proposed methods in Section IV. The experimental analysis and discussion of the proposed algorithms are detailed in Section V. Section VI concludes this article.

II. Related Work

In this section, the tensor factorization unmixing model and hypergraph learning are introduced. The former is the basic model, and the latter is the theoretical basis for model improvement.

A. Tensor Factorization Unmixing

A tensor is a multidimensional form of data storage, and the dimensions of data are called the order of tensors. It can be viewed as the generalization of vectors and matrices in multidimensional space, i.e., vectors can be viewed as 1-D tensors, and matrices can be viewed as 2-D tensors. The definitions are introduced in Table I.

In 2007, Veganzones et al. [40] proposed a BTD, and its variants have achieved a good performance in hyperspectral image unmixing. The BTD decomposes an N-order tensor into R tensor components, and each tensor component is the product of the core tensor and the factor matrix. Given a hyperspectral image $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$, its BTD is expressed as

$$\mathcal{X} = \sum_{r=1}^{R} \Re_r \times_1 \mathbf{A}_r \times_2 \mathbf{B}_r \times_3 \mathbf{C}_r. \tag{1}$$

According to the LSMM, the matrix-vector tensor factorization (MV-NTF) model is obtained by specializing in the BTD model [30]. The MV-NTF unmixing model decomposes a three-order hyperspectral image into the sum of a series of tensor components, and each tensor component is defined as the outer product of a matrix $\mathbf{E}_r$ and a vector $\mathbf{c}_r$, where the matrix $\mathbf{E}_r$ is the product of $\mathbf{A}_r$ and $\mathbf{B}_r$, i.e.,

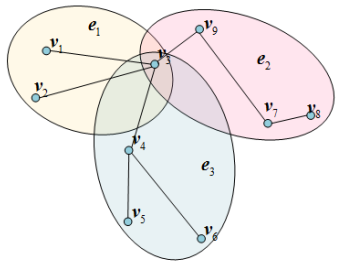
$$\mathcal{X} = \sum_{r=1}^{R} \mathbf{A}_r \cdot \mathbf{B}_r^T \circ \mathbf{c}_r = \sum_{r=1}^{R} \mathbf{E}_r \circ \mathbf{c}_r \tag{2}$$

where $\mathbf{c}_r$ can be regarded as the rth endmember, and $\mathbf{E}_r$ is the corresponding abundance map. This model can be connected with the LSMM and can effectively represent the physical mechanism of unmixing.

B. Hypergraph Learning

A hypergraph can simulate higher order relationships between objects. Given a set of finite elements $\mathbf{V} = \{v_1, v_2, \ldots, v_N\}$, $\varepsilon = \{e_1, e_2, \ldots, e_M\}$ is the subset of V, $\mathbf{G} = (\mathbf{V}, \varepsilon)$ is a hypergraph defined on $\mathbf{V}$ [see Fig. 1(a)], and the elements of $\mathbf{V}$ are called vertices and $e_i \in \varepsilon$ is the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.



(a)

	e_1	e_2	e_3
v_1	1	0	0
v_2	1	0	0
v_3	1	1	1
v_4	0	0	1
v_5	0	0	1
v_6	0	0	1
v_7	0	1	0
v_8	0	1	0
v_9	0	1	0

(b)

Fig. 1. (a) Hypergraph and (b) incidence matrix. Solid nodes represent vertices, and elliptically labeled sets of nodes represent hyperedges.

TABLE I
BASIC NOTATIONS AND THEIR DESCRIPTIONS

Notation	Descriptions
x, x, $\mathbf{X}$, $\mathcal{X}$	Constant, vector, matrix, and Tensor
$\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_n \times \cdots \times I_N}$	Nth-order tensor
$X_{(n)} \in R^{I_n \times (I_1 I_2 \times \ldots I_{n-1} I_{n+1} \ldots I_N)}$	Mode-n unfolding
$(\mathcal{X} \circ \mathcal{Y})_{i_1 i_2 \ldots i_N j_1 j_2 \ldots j_M} = x_{i_1 i_2 \ldots i_N} y_{j_1 j_2 \ldots j_M}$	Out product of two tensors
$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1J}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2J}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{I1}\mathbf{B} & a_{I2}\mathbf{B} & \cdots & a_{IJ}\mathbf{B} \end{bmatrix} \in \mathbb{R}^{IM \times JN}$	Kronecker product of two matrices
$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} \boldsymbol{a}_1 \otimes \boldsymbol{b}_1 & \boldsymbol{a}_2 \otimes \boldsymbol{b}_2 & \cdots & \boldsymbol{a}_K \otimes \boldsymbol{b}_K \end{bmatrix} \in \mathbb{R}^{IJ \times K}$	Khatri-Rao product of two matrices
$\mathbf{A} \overline{\odot} \mathbf{B} = \begin{bmatrix} \mathbf{A}_1 \otimes \mathbf{B}_1 & \mathbf{A}_2 \otimes \mathbf{B}_2 & \cdots & \mathbf{A}_N \otimes \mathbf{B}_N \end{bmatrix}$	partition wise Khatri-Rao product of two matrix

hyperedge of $\mathbf{G}$. The relation between vertices and hyperedges is expressed by an incidence matrix $\mathbf{H} \in R^{|\mathbf{V}| \times |\varepsilon|}$ [see Fig. 1(b)]

$$\mathbf{H}(v_i, \boldsymbol{e}_i) = \begin{cases} 1, & v_i \in \boldsymbol{e}_i \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

For the hyperspectral image $\mathbf{Y}$, each pixel y_i is an element of the vertex set $\mathbf{V}$. Then, each vertex acts as the centroid of a local neighborhood, and the corresponding hyperedge is constructed by the vertex and its k nearest spatial neighbors. Commonly, each hyperedge $\boldsymbol{e}_i \in \varepsilon$ is assigned a positive weight $\omega(\boldsymbol{e}_i)$ to represent the similarity of pixels in the spatial neighborhood, and the weight can be calculated as

$$\omega(e_i) = \sum_{y_j \in e_i} \exp\left(-\frac{\|y_j - y_i\|^2}{\gamma^2}\right) \tag{4}$$

$$\gamma = \frac{1}{KN} \sum_{i=1}^{N} \sum_{\{y_j \in e_i\}} \|y_i - y_j\| \tag{5}$$

where K represents the number of nearest spatial neighbors. The edge degree $\delta(\boldsymbol{e}_i)$ of each hyperedge $\boldsymbol{e}_i \in \varepsilon$ is defined as the number of vertices contained in each hyperedge

$$\delta(\boldsymbol{e}_i) = \sum_{j=1}^{N} \mathbf{H}(v_j, \boldsymbol{e}_i). \tag{6}$$

The vertex degree $d(v_i)$ of each vertex $v_i \in \mathbf{V}$ is defined as the sum of the weights associated with the hyperedge to which the vertex belongs

$$d(v_i) = \sum_{j=1}^{M} \omega(\boldsymbol{e}_j) \mathbf{H}(v_i, \boldsymbol{e}_j). \tag{7}$$

The diagonal matrices $\mathbf{D}_v$ and $\mathbf{D}_e$ are used to represent the degree matrices of vertices and hyperedges, respectively, where the elements on the diagonal are the degrees of each vertex or hyperedge, i.e., $d(v_i)$ or $\delta(\boldsymbol{e}_i)$. Let the diagonal matrix $\mathbf{W}$ represent the weight matrix of the hyperedge, where the elements on the diagonal represent the weight of each hyperedge $\omega(\boldsymbol{e}_i)$. The adjacent matrix $\mathbf{A}$ of $G = (V, \varepsilon, W)$ is defined as $\mathbf{A} = \mathbf{H}^T \mathbf{W} (\mathbf{D}_e)^{-1} \mathbf{H}$. The adjacency matrix represents the adjacent relationship between each vertex. Finally, the unnormalized hypergraph Laplacian matrix is $\mathbf{L}^{\text{Hyper}} = \mathbf{D}_v - \mathbf{A}$.

III. PROPOSED METHOD

As mentioned above, MV-NTF for hyperspectral unmixing can maintain the structural characteristics of hyperspectral images, and this decomposition unmixing model conforms to the physical mechanism of spectral unmixing. However, it fails to take into consideration the spatial–spectral prior to the hyperspectral image and the sparsity of abundances. In this section, the AHGNTF algorithm and AHGMLSTF algorithm are proposed to overcome these limitations.

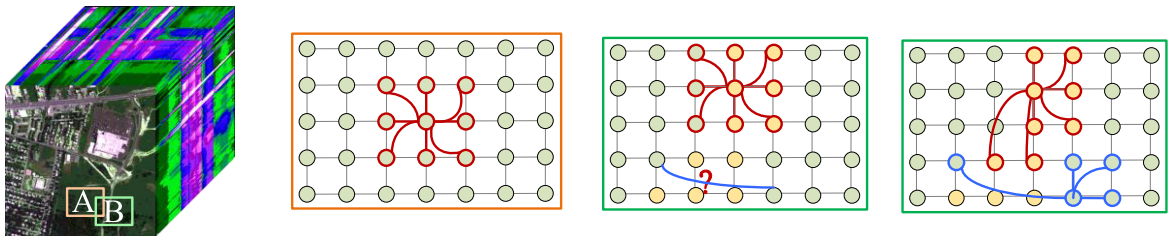
A. Adaptive Hypergraph Regularized Nonnegative Tensor Factorization

The MV-NTF unmixing algorithm decomposes the hyperspectral image tensor into the sum of several tensor components, and each tensor component is the outer product of the matrix and the vector. The objective function can be expressed as

$$\min_{\mathbf{A}_r, \mathbf{B}_r, \mathbf{c}_r} \frac{1}{2} \left\| \mathcal{Y} - \sum_{r=1}^{R} \mathbf{A}_r \cdot \mathbf{B}_r^T \circ \mathbf{c}_r \right\|_F^2 + \frac{1}{2} \left\| \sum_{r=1}^{R} \mathbf{A}_r \mathbf{B}_r^T - 1_{I \times J} \right\|_F^2 \quad \text{s.t. } \mathbf{A}_r \geq 0,\ \mathbf{B}_r \geq 0,\ \mathbf{c}_r \geq 0 \tag{8}$$

where $\mathcal{Y}$ denotes the hyperspectral image with a size of $I \times J$ pixels and K bands. R represents the number of endmembers. $\mathbf{c}_r$ represents the rth endmember, and $\mathbf{A}_r \mathbf{B}_r^T$ denotes its corresponding abundance map, which can be made up of two matrices $\mathbf{A}_r \in \mathbb{R}^{I \times M}$ and $B_r \in \mathbb{R}^{M \times J}$ with $M \leq \min(I, J)$. The second term indicates the sum-to-one constraint, which means that the sum of the abundances in each pixel is one. $1_{I \times J}$ denotes an $I \times J$ matrix with all entries are one. According to the physical mechanism of spectral unmixing, the sum of abundance fractions in each pixel is one, and the endmember and abundance matrices are nonnegative.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 2. Hypergraph construction: (a) A: neighborhood with the same endmember and B: neighborhood with different endmembers; (b) k nearest neighbor method; (c) limitation of the k nearest neighbor method; and (d) proposed hypergraph construction method.

In order to exploit the local spatial priors in a hyperspectral image, graphs are usually used to construct pairwise relationships between pixels. To better gauge the relationships between pixels in a local area, hypergraphs are used to represent multivariate relationships. According to hypergraph learning, spatial structure is incorporated into the tensor factorization model as a regularization term to represent the consistency of abundances within the same hyperedge.

To construct hypergraphs in hyperspectral unmixing, pixels are generally used as vertices of hypergraphs, and the corresponding hyperedge is constructed by the vertex and its k nearest neighbors. Typically, the Euclidean distance is used to calculate the distance between two pixels, and each hyperedge is assigned a weight that represents the similarity of the pixels contained in the local neighborhood. This method can effectively utilize local spatial information but has its limitations at the edge of two dissimilar objects [see Fig. 2(c)]. This motivates hypergraph construction in the proposed method.

Given a hyperspectral image $\mathbf{Y} \in \mathbb{R}^{L\times N}$ with N pixels and L bands, each pixel is viewed as a vertex, and a hyperedge consists of a vertex and its k nearest neighbors. When constructing the k nearest neighbors, the SAD is used to calculate the distance between pixels as

$$\mathrm{SAD}(\boldsymbol{y}_i, \boldsymbol{y}_j) = \arccos\left(\frac{\boldsymbol{y}_i^T \boldsymbol{y}_j}{\|\boldsymbol{y}_i\| \cdot \|\boldsymbol{y}_j\|}\right). \tag{9}$$

To represent the spectral similarity of different pixels, the incidence matrix of the hyperedge is modified as

$$\mathbf{H}(v_i, \boldsymbol{e}_i) = \begin{cases} \mathrm{SAD}(v_i, \boldsymbol{e}_i), & v_i \in \boldsymbol{e}_i \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Accordingly, a vertex v_i is assigned to $\boldsymbol{e}_i$ based on the SAD.

Next, each hyperedge $\boldsymbol{e}_i \in \varepsilon$ is assigned a weight $\omega(\boldsymbol{e}_i)$ that represents the similarity of the pixels in the neighborhood

$$\omega(\boldsymbol{e}_i) = \sum_{\boldsymbol{y}_j \in \boldsymbol{e}_i} \exp\left(-\frac{\|\boldsymbol{y}_j - \boldsymbol{y}_i\|^2}{\gamma^2}\right) \tag{11}$$

$$\gamma = \frac{1}{KN} \sum_{i=1}^{N} \sum_{\{\boldsymbol{y}_j \in \boldsymbol{e}_i\}} \|\boldsymbol{y}_i - \boldsymbol{y}_j\|. \tag{12}$$

This hypergraph construction method utilizes spatial and spectral priors simultaneously, and the hypergraph can overcome the limitation in Fig. 2(c). The Laplacian matrix of a hypergraph is defined as

$$\mathbf{L}^{\mathrm{Hyper}} = \mathbf{D}_v - \mathrm{HW}(\mathbf{D}_{\mathbf{e}})^{-1}(\mathbf{H})^T. \tag{13}$$

Adding the hypergraph regularization term to the tensor decomposition unmixing model and omitting the superscript (8) can be written as

$$\begin{aligned} \min_{\mathbf{A},\mathbf{B},\mathbf{c}_r} & \frac{1}{2}\|\mathcal{Y} - \sum_{r=1}^{R}\left(\mathbf{A}_r\mathbf{B}_r^T\right)\circ\mathbf{c}_r\|_F^2 + \frac{1}{2}\|1_R\left(\mathbf{A}\mathbf{B}^T\right)^T - 1_{I\times J}\|_F^2 \\ & + \gamma\,\mathrm{tr}\left(\left(\mathbf{A}\mathbf{B}^T\right)^T\mathbf{L}^{\mathrm{Hyper}}\left(\mathbf{A}\mathbf{B}^T\right)\right) \\ \text{s.t. } & \mathbf{A} \geq 0,\ \ \mathbf{B} \geq 0,\ \ \mathbf{c}_r \geq 0 \end{aligned} \tag{14}$$

where $\mathbf{A}\mathbf{B}^T = [\mathbf{A}_1\mathbf{B}_1^T, \ldots, \mathbf{A}_r\mathbf{B}_r^T, \ldots, \mathbf{A}_R\mathbf{B}_R^T] \in \mathbb{R}^{IJ\times R}$ denotes the abundance matrix containing the abundance fractions of all pixels. $\mathbf{L}^{\mathrm{Hyper}} \in \mathbb{R}^{IJ\times IJ}$ represents the Laplacian matrix of the hypergraph.

Another factor that affects the performance of hypergraph regularized hyperspectral unmixing is the size of k nearest neighbors of the hypergraph. In the proposed method AHGNTF, a group of hyperedges with different sizes of nearest neighbors with a certain range. Hyperedges with different neighbors have different effects in hyperspectral unmixing, for example, when some pixels in the hyperedge are at the edge of different features. In this case, the hyperedge may have a little or negative impact on the unmixing performance. In the proposed algorithm, the hyperedge weights of k neighborhoods are updated, and more efficient hyperedges are selected by imposing the sparse constraint on the weights. The adaptive hypergraph construction is illustrated in Fig. 2(d). Equation (14) can be rewritten as

$$\begin{aligned} \min_{\mathbf{A},\mathbf{B},\mathbf{c}_r,W} & \frac{1}{2}\|\mathcal{Y} - \sum_{r=1}^{R}\left(\mathbf{A}_r\mathbf{B}_r^T\right)\circ\mathbf{c}_r\|_F^2 + \frac{1}{2}\|1_R\left(\mathbf{A}\mathbf{B}^T\right)^T - 1_{I\times J}\|_F^2 \\ & + \gamma\,\mathrm{tr}\left(\left(\mathbf{A}\mathbf{B}^T\right)^T\mathbf{L}^{\mathrm{Hyper}}\left(\mathbf{A}\mathbf{B}^T\right)\right) + \delta\|\mathbf{W}\|_1 \\ \text{s.t. } & \mathbf{A} \geq 0,\quad \mathbf{B} \geq 0,\ \ \mathbf{c}_r \geq 0,\ \ \mathbf{W} \geq 0 \end{aligned} \tag{15}$$

where the first two terms represent the error between the estimated image and the original image and sum-to-one constraint, the third term represents the hypergraph regularization term, and the fourth term represents the weights among hyperedges. The weight matrix satisfies the sum-to-one and nonnegativity constraints. By updating the regularization terms, the adaptive hypergraph can be obtained, which can better represent the spatial–spectral information in a local neighborhood.

The ADMM is adopted to solve the objective function [49]. The objective function is divided into several subproblems, which are solved in an iterative process. Fix the other variables

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

while solving for one of them during the iteration. By introducing auxiliary variables, (15) becomes as follows:

$$\begin{aligned}\min_{\mathbf{U}_1,\mathbf{V}_1,\mathbf{U}_2,\mathbf{V}_2}\ &\frac{1}{2}\|\mathcal{Y}-\sum_{r=1}^{R}\mathbf{A}_r\mathbf{B}_r^T\circ\mathbf{c}_r\|_F^2+\frac{1}{2}\|1_R\left(\mathbf{A}\mathbf{B}^T\right)^T-1_{I\times J}\|_F^2\\&+\gamma\mathrm{tr}\Big(\left(\mathbf{U}_1\mathbf{V}_1^T\right)^T\mathbf{L}^{\mathrm{Hyper}}\left(\mathbf{U}_1\mathbf{V}_1^T\right)\Big)+\delta\|\mathbf{W}\|_1\\&\text{s.t. }\mathbf{U}_1-\mathbf{A}=0,\mathbf{V}_1-\mathbf{B}=0.\end{aligned}\tag{16}$$

The augmented Lagrangian form of (16) is obtained in the following equation:

$$\begin{aligned}\boldsymbol{L}=&\min_{\mathbf{A}_r,\mathbf{B}_r,\mathbf{c}_r,\mathbf{U}_1,\mathbf{V}_1,\mathbf{W}}\frac{1}{2}\|\mathcal{Y}-\sum_{r=1}^{R}\mathbf{A}_r\mathbf{B}_r^T\circ\mathbf{c}_r\|_F^2\\&+\frac{1}{2}\|\mathbf{1}_R\left(\mathbf{A}\mathbf{B}^T\right)^T-1_{I\times J}\|_F^2\\&+\delta\|\mathbf{W}\|_1+\gamma\mathrm{tr}\Big(\left(\mathbf{U}_1\mathbf{V}_1^T\right)^T\mathbf{L}^{\mathrm{Hyper}}\left(\mathbf{U}_1\mathbf{V}_1^T\right)\Big)\\&+\mathrm{tr}\left[\mathbf{D}_1^T(\mathbf{U}_1-\mathbf{A})\right]\\&+\mathrm{tr}\left[\mathbf{D}_2^T(\mathbf{V}_1-\mathbf{A})\right]+\frac{\mu}{2}\left(\|\mathbf{A}-\mathbf{U}_1\|_F^2+\|\mathbf{B}-\mathbf{V}_1\|_F^2\right)\end{aligned}\tag{17}$$

where $\mathbf{D}_1$ and $\mathbf{D}_2$ are the Lagrange multipliers, and μ is the penalty parameter that controls the relationship between the original variables and the auxiliary variables. Then, (17) can be rewritten in an equivalent form

$$\begin{aligned}\boldsymbol{L}=&\min_{\mathbf{A}_r,\mathbf{B}_r,\mathbf{c}_r,\mathbf{U}_1,\mathbf{V}_1,\mathbf{W}}\frac{1}{2}\|\mathcal{Y}-\sum_{r=1}^{R}\mathbf{A}_r\mathbf{B}_r^T\circ\mathbf{c}_r\|_F^2\\&+\frac{1}{2}\|\mathbf{1}_R\left(\mathbf{A}\mathbf{B}^T\right)^T-1_{I\times J}\|_F^2\\&+\delta\|\mathbf{W}\|_1+\gamma\mathrm{tr}\Big(\left(\mathbf{U}_1\mathbf{V}_1^T\right)^T\mathbf{L}^{\mathrm{Hyper}}\left(\mathbf{U}_1\mathbf{V}_1^T\right)\Big)\\&+\frac{\mu}{2}\left(\|\mathbf{A}-\mathbf{U}_1-\boldsymbol{\Lambda}_1\|_F^2+\|\mathbf{B}-\mathbf{V}_1-\boldsymbol{\Lambda}_2\|_F^2\right)\end{aligned}\tag{18}$$

where $\boldsymbol{\Lambda}_i=(1/\mu)\mathbf{D}_i, i=1,2$ are the scaled Lagrange multipliers.

Update A: The tensor form of the hyperspectral image is unfolded in the mode-I way, the items irrelevant to $\mathbf{A}$ are removed, and the subproblem of solving $\mathbf{A}$ becomes

$$\begin{aligned}f(\mathbf{A})=&\min_{\mathbf{A}}\frac{1}{2}\|\mathbf{Y}_{JK\times I}-\left(\mathbf{B}\overline{\odot}\mathbf{C}\right)\cdot\mathbf{A}^T\|_F^2\\&+\frac{1}{2}\|\mathbf{1}_R\left(\mathbf{A}\mathbf{B}^T\right)^T-1_{I\times J}\|_F^2\\&+\frac{\mu}{2}\left(\|\mathbf{A}-\mathbf{U}_1-\boldsymbol{\Lambda}_1\|_F^2\right).\end{aligned}\tag{19}$$

The partial derivative of $\mathbf{A}$ is calculated to obtain the update rule of $\mathbf{A}$

$$\begin{aligned}\mathbf{A}\leftarrow&\mathbf{A}.*\left(\mathbf{Y}_{JK\times I}^T\mathbf{S}+1_{I\times J}\mathbf{B}+\mu(\mathbf{U}_1+\boldsymbol{\Lambda}_1)\right)\\&./\left(\mathbf{A}\mathbf{S}^T\mathbf{S}+\mathbf{A}\mathbf{B}^T\mathbf{B}+\mu\mathbf{A}\right)\end{aligned}\tag{20}$$

$$\mathbf{S}=\mathbf{B}\overline{\odot}\mathbf{C}.\tag{21}$$

Update B: The tensor form of the hyperspectral image is unfolded in the mode-II way, the items irrelevant to $\mathbf{B}$ are removed, and the subproblem of solving $\mathbf{B}$ becomes

$$\begin{aligned}f(\mathbf{B})=&\min_{\mathbf{B}}\frac{1}{2}\|\mathbf{Y}_{KI\times J}-\left(\mathbf{C}\overline{\odot}\mathbf{A}\right)\cdot\mathbf{B}^T\|_F^2\\&+\frac{1}{2}\|\mathbf{1}_R\left(\mathbf{A}\mathbf{B}^T\right)^T-1_{I\times J}\|_F^2\\&+\frac{\mu}{2}\left(\|\mathbf{B}-\mathbf{V}_1-\boldsymbol{\Lambda}_2\|_F^2\right).\end{aligned}\tag{22}$$

The partial derivative of $\mathbf{B}$ is calculated to obtain the update rule of $\mathbf{B}$

$$\begin{aligned}\mathbf{B}\leftarrow&\mathbf{B}.*\left(\mathbf{Y}_{KI\times J}^T\mathbf{S}+\mathbf{1}_{I\times J}^T\mathbf{A}+\mu(\mathbf{V}_1+\boldsymbol{\Lambda}_2)\right)\\&./\left(\mathbf{B}\mathbf{S}^T\mathbf{S}+\mathbf{B}\mathbf{A}^T\mathbf{A}+\mu\mathbf{B}\right)\end{aligned}\tag{23}$$

$$\mathbf{S}=\mathbf{C}\overline{\odot}\mathbf{A}.\tag{24}$$

Update C: The tensor form of the hyperspectral image is unfolded in the mode-III way, and the subproblem of solving $\mathbf{C}$ becomes

$$\begin{aligned}f(\mathbf{C})=\min_{\mathbf{C}}\frac{1}{2}\Big\|\mathbf{Y}_{IJ\times K}-\big[&(\mathbf{A}_1\odot\mathbf{B}_1)\mathbf{1}_{\mathbf{L}_1}\\&\cdots(\mathbf{A}_N\odot\mathbf{B}_N)\mathbf{1}_{\mathbf{L}_N}\big]\cdot\mathbf{C}^{\mathbf{T}}\Big\|_F^2.\end{aligned}\tag{25}$$

The partial derivative of $\mathbf{C}$ is calculated to obtain the update rule of $\mathbf{C}$

$$\mathbf{C}\leftarrow\mathbf{C}.*\mathbf{Y}_{IJ\times K}^T\mathbf{S}./(\mathbf{C}\mathbf{S}^T\mathbf{S})\tag{26}$$

$$\mathbf{S}=[(\mathbf{A}_1\odot\mathbf{B}_1)1_L,\ldots,(\mathbf{A}_R\odot\mathbf{B}_R)1_L].\tag{27}$$

Update $\mathbf{U}_1$ and $\mathbf{V}_1$: The subproblems of solving $\mathbf{U}_1$ and $\mathbf{V}_1$ can be written as

$$f(\mathbf{U}_1)=\gamma\mathrm{tr}\left(\left(\mathbf{U}_1\mathbf{V}_1^{\mathbf{T}}\right)^{\mathbf{T}}\mathbf{L}^{\mathrm{Hyper}}\mathbf{U}_1\mathbf{V}_1^{\mathbf{T}}\right)+\frac{\mu}{2}\|\mathbf{A}-\mathbf{U}_1-\boldsymbol{\Lambda}_1\|_F^2\tag{28}$$

$$f(\mathbf{V}_1)=\gamma\mathrm{tr}\left(\left(\mathbf{U}_1\mathbf{V}_1^{\mathbf{T}}\right)^{\mathbf{T}}\mathbf{L}^{\mathrm{Hyper}}\mathbf{U}_1\mathbf{V}_1^{\mathbf{T}}\right)+\frac{\mu}{2}\|\mathbf{B}-\mathbf{V}_1-\boldsymbol{\Lambda}_2\|_F^2.\tag{29}$$

The update rules of $\mathbf{U}_1$ and $\mathbf{V}_1$ are calculated as

$$\mathbf{U}_1=\mu(A-\boldsymbol{\Lambda}_1)(2\gamma\mathbf{L}^{\mathbf{Hyper}}+\mu\mathbf{I})^{-1}\tag{30}$$

$$\mathbf{V}_1=\mu(B-\boldsymbol{\Lambda}_2)(2\gamma\mathbf{L}^{\mathbf{Hyper}}+\mu\mathbf{I})^{-1}.\tag{31}$$

Update W: Since $\mathbf{L}^{\mathrm{Hyper}}=\mathbf{D}_v-\mathbf{H}\mathbf{W}(\mathbf{D}_e)^{-1}(\mathbf{H})^T$, the subproblem of $\mathbf{W}$ can be written as

$$f(\mathbf{W})=\gamma\mathrm{tr}\Big(-\left(\mathbf{U}_1\mathbf{V}_1^T\right)^T\left(\mathbf{H}\mathbf{W}(\mathbf{D}_e)^{-1}(\mathbf{H})^T\right)\mathbf{U}_1\mathbf{V}_1^T\Big)+\delta\|\mathbf{W}\|_1.\tag{32}$$

Since $\mathbf{W}$ and $\mathbf{D}_e^{-1}$ are the diagonal matrices, let $\mathbf{R}=\mathbf{U}_1\mathbf{V}_1\mathbf{H}$, the first term becomes

$$\begin{aligned}&-\gamma\mathrm{tr}\left(\mathbf{R}^T\mathbf{W}(\mathbf{D}_e)^{-1}\mathbf{R}\right)\\&=-\gamma\left(\mathbf{W}_1\times(\mathbf{R}_1)^2\times\mathbf{D}_{e1}^{-1}+\cdots+\mathbf{W}_m\times(\mathbf{R}_m)^2\times\mathbf{D}_{em}^{-1}\right).\end{aligned}\tag{33}$$

Formula (33) can be written as

$$\begin{aligned}f(\mathbf{W})=&-\gamma\left(\mathbf{W}_1\times(\mathbf{R}_1)^2\times\mathbf{D}_{e1}^{-1}+\cdots+\mathbf{W}_m\times(\mathbf{R}_m)^2\times\mathbf{D}_{em}^{-1}\right)\\&+\delta\|\mathbf{W}\|_1.\end{aligned}\tag{34}$$

The coordinate descent algorithm is adopted to solve the problem.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Adaptive Hypergraph Regularized Nonnegative Tensor Factorization AHGNTF

input: $\mathbf{Y}$, γ, μ, δ, initialization $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{U}_1$, $\mathbf{V}_1$
Repeat
Construct the hypergraph Laplacian $\mathbf{L}^{\text{Hyper}}$ using (68)
Update $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ by Equation (20) (23) (26)
Update $\mathbf{U}_1$ and $\mathbf{V}_1$ by Equation (28)(29)
Update $\mathbf{\Lambda}_1$ and $\mathbf{\Lambda}_2$ by Equation (35)(36)
Update weight $\mathbf{W}$ with iterative coordinate descent method.
After obtaining $\mathbf{W}$, update $\mathbf{L}^{\text{Hyper}}$
Until convergence
Output: abundance matrix $\mathbf{AB}^T$ endmember matrix $\mathbf{C}$

Update $\mathbf{\Lambda}$*:* The updating rule of the scaled Lagrange multipliers is obtained by gradient ascent, where t represents the number of iterations

$$\mathbf{\Lambda}_1^t = \mathbf{\Lambda}_1^{t-1} - \mathbf{A}^t + \mathbf{U}_1^t \tag{35}$$

$$\mathbf{\Lambda}_2^t = \mathbf{\Lambda}_2^{t-1} - \mathbf{B}^t + \mathbf{V}_1^t. \tag{36}$$

The nonnegative constraint is directly imposed in the iterative process, and the indicator function guarantees the nonnegativity of abundance by setting the negative value to 0 during iteration. There are two iteration stopping criteria. Either the maximum number of iterations is 3000, or the gradient difference of the objective function (18) between the current and the previous iterations is less than 1e-4, the iteration will stop. R is known in the simulated dataset and in the hyperspectral image with reference endmember matrix, in the real hyperspectral scene, R can be obtained using the virtual dimensionality (VD) method [50]. $\mathbf{E} = \mathbf{AB}^T$ and $\mathbf{C}$ represent the abundance and endmember matrices, respectively, which can be obtained by VCA and full constrained least-square (FCLS) algorithms as initial matrices [8], [10]. Then, $\mathbf{A}$ and $\mathbf{B}$ are initialized by $\mathbf{E}$ using the NMF algorithm. With obtained A and B, the auxiliary variables are set to: $\mathbf{U}_1 = A$ and $\mathbf{V}_1 = B$. The proposed algorithm is summarized in Algorithm 1.

According to Algorithm1, the main time cost spent on updating $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ has given in (20), (23), and (26), and the computational complexity analysis of these variables is as follows: the number of floating-point operations of update $\mathbf{A}$ is $R(6I + 2IJK + 2JKR + 2IR + JK + 2JR + 2IJ)$, the number of floating-point operations of update $\mathbf{B}$ is $R(6J + 2IJK + 2IKR + 2JR + IK + 2IJ + 2IR)$, and the number of floating-point operations of update $\mathbf{C}$ is $R(2K + 2IJK + 2IJR + 2KR + 3IJ)$. Therefore, after t iterations, the algorithm complexity is $\mathcal{O}(tIJKR + tIKR^2 + tJKR^2)$, where R is the number of endmembers of a hyperspectral image, I and J are the row number and column number of the image, respectively, and K is the number of bands.

B. Adaptive Hypergraph Regularized Multilayer Sparse Tensor Factorization

The concept of multilayer is proposed to reduce the risk of being trapped in local minima and improve the performance of matrix factorization. In this method, this concept is used to solve tensor factorization in hyperspectral unmixing.

The mathematical representation of multilayer matrix factorization for hyperspectral unmixing is formulated as follows [51]:

$$\mathbf{Y} = \mathbf{X}_1\mathbf{S}_1,\ \mathbf{S}_1 = \mathbf{X}_2\mathbf{S}_2, \ldots, \mathbf{S}_{l-1} = \mathbf{X}_l\mathbf{S}_l, \ldots, \mathbf{S}_{L-1} = \mathbf{X}_L\mathbf{S}_L \tag{37}$$

where $\mathbf{Y}$ represents the hyperspectral image, and $\mathbf{X}_l$ and $\mathbf{S}_l$ represent the endmember matrix and abundance matrix in the lth layer, respectively. $Y = \mathbf{X}_1\mathbf{X}_2, \ldots, \mathbf{X}_l, \ldots, \mathbf{X}_L\mathbf{S}_L$ can be represented by multilayer endmember and abundance matrices, the obtained abundance matrix is $\mathbf{S}_L$, and the obtained endmember matrix $\mathbf{X} = \mathbf{X}_1\mathbf{X}_2, \ldots, \mathbf{X}_l, \ldots, \mathbf{X}_L$ is the product of multilayer endmember matrix.

It can be observed that most of the pixels in the image are mixed by a small number of endmembers, the abundance matrix has sparse property, and the l_1-norm is imposed on the abundance to indicate abundance sparsity. Considering the sparsity of abundance in each layer after decomposition, a sparsity constraint is imposed on the abundance matrix. The real endmember matrix does not have sparseness, but due to the multilayer decomposition, each layer of the endmember matrix is sparse, so the sparse representation of the endmember is forced in each layer; the main idea is to model the endmember by cascade connections of sparse matrices. Such multilayer factorization is biologically motivated and extensive experiments have verified that such sparseness improves NTF performance [52]. The objective function is formulated as

$$\begin{aligned}\min_{\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{W}} &\|\mathcal{Y}_l - \left(\mathbf{AB}^T\right)_l \circ \mathbf{C}_l\|_F^2 + \delta\|\mathbf{W}\|^2 + \lambda\|\left(\mathbf{AB}^T\right)_l\|_1 \\ &+ \beta\|\mathbf{C}_l\|_1 + \gamma\text{tr}\Big(\left(\mathbf{AB}^T\right)_l^T \mathbf{L}^{\text{Hyper}}\left(\mathbf{AB}^T\right)_l\Big) \\ \text{s.t. } &\mathbf{U}_1 = \mathbf{A}_l,\ \mathbf{V}_1 = \mathbf{B}_l,\ \mathbf{U}_2 = \mathbf{A}_l,\ \mathbf{V}_2 = \mathbf{B}_l,\ \mathbf{U}_3 = \mathbf{C}_l\end{aligned} \tag{38}$$

where $\lambda = \lambda_0 e^{-t/\tau}$, t is the number of iterations, and λ_0 and τ are constants constraining the effect of sparsity. This regularization term prevents overfitting. To increase the effect of sparsity constraint on abundance, according to experience, set $\lambda = 2\beta$ in the experiments. Equation (38) can be written in an augmented Lagrangian form as

$$\begin{aligned}\boldsymbol{L} = \min_{\mathbf{A}_l,\mathbf{B}_l,\mathbf{C}_l,\mathbf{U}_1,\mathbf{V}_1,\mathbf{U}_2,\mathbf{V}_2,\mathbf{U}_3} &\frac{1}{2}\|\mathcal{Y}_l - \mathbf{A}_l\mathbf{B}_l^T \circ \mathbf{C}_l\|_F^2 + \lambda\|\mathbf{U}_2\mathbf{V}_2^T\|_1 \\ &+ \delta\|\mathbf{W}\|_1 + \beta\|\mathbf{U}_3\|_1 + \gamma\text{tr}\Big(\left(\mathbf{U}_1\mathbf{V}_1^T\right)^T \mathbf{L}^{\text{Hyper}}\left(\mathbf{U}_1\mathbf{V}_1^T\right)\Big) \\ &+ \text{tr}\left[\mathbf{D}_1^T(\mathbf{U}_1 - \mathbf{A}_l)\right] + \text{tr}\left[\mathbf{D}_2^T(\mathbf{V}_1 - \mathbf{B}_l)\right] + \text{tr}\left[\mathbf{D}_3^T(\mathbf{U}_2 - \mathbf{A}_l)\right] \\ &+ \text{tr}\left[\mathbf{D}_4^T(\mathbf{V}_2 - \mathbf{B}_l)\right] + \text{tr}\left[\mathbf{D}_5^T(\mathbf{U}_3 - \mathbf{C}_l)\right] \\ &+ \frac{\mu}{2}\begin{pmatrix}\|\mathbf{A}_l - \mathbf{U}_1\|_F^2 + \|\mathbf{B}_l - \mathbf{V}_1\|_F^2 + \|\mathbf{A}_l - \mathbf{U}_2\|_F^2 \\ + \|\mathbf{B}_l - \mathbf{V}_2\|_F^2 + \|\mathbf{C}_l - \mathbf{U}_3\|_F^2\end{pmatrix}\end{aligned} \tag{39}$$

where $\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3, \mathbf{D}_4, \mathbf{D}_5$ are the Lagrange multipliers. Let $\mathbf{U} = [\mathbf{U}_1 \quad \mathbf{U}_2]^T$ and $\mathbf{V} = [\mathbf{V}_1 \quad \mathbf{V}_2]^T$. Equation (39) can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

simplified to

$$\begin{aligned}\boldsymbol{L} = \min_{\mathbf{A}_l,\mathbf{B}_l,\mathbf{C}_l,\mathbf{U}_1,\mathbf{V}_1,\mathbf{U}_2,\mathbf{V}_2,\mathbf{U}_3} & \frac{1}{2}\|\mathcal{Y}_l - \mathbf{A}_l\mathbf{B}_l^T \circ \mathbf{C}_l\|_F^2 + \lambda\|\mathbf{U}_2\mathbf{V}_2^T\|_1 \\ & + \gamma \mathrm{tr}\Big(\big(\mathbf{U}_1\mathbf{V}_1^T\big)^T \mathbf{L}^{\mathrm{Hyper}}\big(\mathbf{U}_1\mathbf{V}_1^T\big)\Big) + \delta\|\mathbf{W}\|_1 + \beta\|\mathbf{U}_3\|_1 \\ & + \frac{\mu}{2}\big(\|\mathbf{A}_l - \mathbf{U} + \Lambda\|_F^2 + \|\mathbf{B}_l - \mathbf{V} + \Lambda\|_F^2 \\ & \quad + \|\mathbf{C}_l - \mathbf{U}_3 + \Lambda\|_F^2\big) \end{aligned} \tag{40}$$

where $\mathbf{\Lambda}_i = (1/\mu)\mathbf{D}_i$, $i = 1, 2, \ldots, 5$ are the scaled Lagrange multipliers.

The update rules for $\mathbf{A}_l$, $\mathbf{B}_l$, and $\mathbf{C}_l$ are similar to the AHGNTF algorithm

$$\mathbf{A}_l \leftarrow \mathbf{A}_l .* \big(Y_{JK\times I}^T\mathbf{S} + \mu(\mathbf{U} + \mathbf{\Lambda})\big)./\mathbf{A}_l(\mathbf{S}^T\mathbf{S} + 2\mu\mathbf{I}) \tag{41}$$

$$\mathbf{S} = \mathbf{B}_l \overline{\odot} \mathbf{C}_l \tag{42}$$

$$\mathbf{B}_l \leftarrow \mathbf{B}_l .* \big(\mathbf{Y}_{KI\times J}^T\mathbf{S} + \mu(\mathbf{V} + \mathbf{\Lambda})\big)./\mathbf{B}_l(\mathbf{S}^T\mathbf{S} + 2\mu\mathbf{I}) \tag{43}$$

$$\mathbf{S} = \mathbf{C}_l \overline{\odot} A_l \tag{44}$$

$$\mathbf{C}_l \leftarrow \mathbf{C}_l .* \big(\mathbf{Y}_{IJ\times K}^T\mathbf{S} + \mu(\mathbf{U}_3 + \mathbf{\Lambda})\big)./\mathbf{C}_l(\mathbf{S}^T\mathbf{S} + 2\mu\mathbf{I}) \tag{45}$$

$$\mathbf{S} = [(\mathbf{A}_{1l} \odot \mathbf{B}_{1l})1_L \cdots (\mathbf{A}_{Rl} \odot \mathbf{B}_{Rl})1_L]. \tag{46}$$

Update $\mathbf{U}_1$ *and* $\mathbf{V}_1$*:* The subproblems of $\mathbf{U}_1$ and $\mathbf{V}_1$ can be written as

$$f(\mathbf{U}_1) = \gamma \mathrm{tr}\Big(\big(\mathbf{U}_1\mathbf{V}_1^T\big)^T \mathbf{L}^{\mathrm{Hyper}}\big(\mathbf{U}_1\mathbf{V}_1^T\big)\Big) + \frac{\mu}{2}\|\mathbf{A}_l - \mathbf{U}_1 + \mathbf{\Lambda}_1\|_F^2 \tag{47}$$

$$f(\mathbf{V}_1) = \gamma \mathrm{tr}\Big(\big(\mathbf{U}_1\mathbf{V}_1^T\big)^T \mathbf{L}^{\mathrm{Hyper}}\big(\mathbf{U}_1\mathbf{V}_1^T\big)\Big) + \frac{\mu}{2}\|\mathbf{B}_l - \mathbf{V}_1 + \mathbf{\Lambda}_2\|_F^2. \tag{48}$$

The update rules are calculated as

$$\mathbf{U}_1 = \mu(\mathbf{A}_l - \Lambda)(2\gamma\mathbf{L}^{\mathbf{Hyper}} + \mu\mathbf{I})^{-1} \tag{49}$$

$$\mathbf{V}_1 = \mu(\mathbf{B}_l - \Lambda)(2\gamma\mathbf{L}^{\mathbf{Hyper}} + \mu\mathbf{I})^{-1}. \tag{50}$$

Update $\mathbf{U}_2$, $\mathbf{U}_3$*, and* $\mathbf{V}_2$*:* The subproblems of $\mathbf{U}_2$, $\mathbf{U}_3$, and $\mathbf{V}_2$ can be written as

$$f(\mathbf{U}_2) = \arg\min \lambda\|\mathbf{U}_2\mathbf{V}_2^T\|_1 + \frac{\mu}{2}\|\mathbf{A}_l - \mathbf{U}_2 - \mathbf{\Lambda}\|_F^2 \tag{51}$$

$$f(\mathbf{V}_2) = \arg\min \lambda\|\mathbf{U}_2\mathbf{V}_2^T\|_1 + \frac{\mu}{2}\|\mathbf{B}_l - \mathbf{V}_2 - \mathbf{\Lambda}_2\|_F^2 \tag{52}$$

$$f(\mathbf{U}_3) = \arg\min \beta\|\mathbf{U}_3\|_1 + \frac{\mu}{2}\|\mathbf{c}_l - \mathbf{U}_3 - \mathbf{\Lambda}\|_F^2. \tag{53}$$

The iterative shrinkage thresholding algorithm is used to solve these subproblems.

Update $\mathbf{\Lambda}_i$*:* The scaled Lagrangian multipliers are updated by performing gradient ascent sequentially, where t is the number of iterations

$$\mathbf{\Lambda}_1^t = \mathbf{\Lambda}_1^{t-1} - \mathbf{A}_l^t + \mathbf{U}_1^t \tag{54}$$

$$\mathbf{\Lambda}_2^t = \mathbf{\Lambda}_2^{t-1} - \mathbf{B}_l^t + \mathbf{V}_1^t \tag{55}$$

$$\mathbf{\Lambda}_3^t = \mathbf{\Lambda}_3^{t-1} - \mathbf{A}_l^t + \mathbf{U}_2^t \tag{56}$$

$$\mathbf{\Lambda}_4^t = \mathbf{\Lambda}_4^{t-1} - \mathbf{B}_l^t + \mathbf{V}_2^t \tag{57}$$

$$\mathbf{\Lambda}_5^t = \mathbf{\Lambda}_5^{t-1} - \mathbf{A}_l^t + \mathbf{U}_3^t. \tag{58}$$

After the endmember $\mathbf{X}_l = \mathbf{C}_l$ and abundance $\mathbf{S}_l = \mathbf{A}_l\mathbf{B}_l^T$ of lth layer are obtained, the error is minimized iteratively

$$\min \frac{1}{2}\|\mathcal{Y} - \mathbf{X}_1\mathbf{X}_2, \ldots, \mathbf{X}_{l-1}\mathbf{X}_l\mathbf{X}_{l+1}, \ldots, \mathbf{X}_L\mathbf{S}_L\|_F^2. \tag{59}$$

Algorithm 2 Adaptive Hypergraph Regularized Multilayer Sparse Tensor Factorization AHGMLSTF

input: $\mathbf{Y}$, γ, μ, λ, β, δ, initialization $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{U}$, $\mathbf{V}$
For each layer
Construct the hypergraph Laplacian $\mathbf{L}^{\mathrm{Hyper}}$
Update $\mathbf{A}_l$, $\mathbf{B}_l$ and $\mathbf{C}_l$ by Equation (41)(43)(45)
Update $\mathbf{U}_1$ and $\mathbf{V}_1$ by Equation (49)(50)
Update $\mathbf{U}_2\mathbf{V}_2$ and $\mathbf{U}_3$ by iterative shrinkage thresholding
Update $\mathbf{\Lambda}_1$, $\mathbf{\Lambda}_2$, $\mathbf{\Lambda}_3$, $\mathbf{\Lambda}_4$, $\mathbf{\Lambda}_5$ by Equation (54)-(58)
Update weight $\mathbf{W}$ with the iterative coordinate descent method.
After obtaining w, update $\mathbf{L}^{Hyper}$
Until convergence
for all layers
update $\mathbf{X}_l$ and $\mathbf{S}_l$ by Equation (63)(64)
Until convergence
Output: abundance matrix $\mathbf{S}$ endmember matrix $\mathbf{X}$

$\mathbf{\Omega}_{l-1}$ and $\tilde{\boldsymbol{S}}_l$ are defined as

$$\mathbf{\Omega}_{l-1} = \mathbf{X}_1\mathbf{X}_2, \ldots, \mathbf{X}_{l-1} \tag{60}$$

$$\tilde{\boldsymbol{S}}_l = \mathbf{X}_l\mathbf{X}_{l+1}, \ldots, \mathbf{X}_L\mathbf{S}_L \tag{61}$$

where $\tilde{\boldsymbol{S}}_l$ denotes the lth layer of the abundance matrix, and the formula becomes

$$\min_{\mathbf{X}_l\tilde{S}_l} \frac{1}{2}\|\mathcal{Y} - \mathbf{\Omega}_{l-1}\mathbf{X}_l\tilde{\mathbf{S}}_l\|_F^2. \tag{62}$$

To obtain the local optimal solution of $\mathbf{X}_l$ and $\tilde{\boldsymbol{S}}_l$, the gradient descent method is used to calculate the updating rule. The endmember and abundance of the lth layer are

$$\mathbf{X}_l = \mathbf{X}_l .* \big(\mathbf{\Omega}_{l-1}^T\mathbf{Y}\tilde{S}_l^T\big)./\big(\mathbf{\Omega}_{l-1}^T\mathbf{\Omega}_{l-1}\mathbf{X}_l\tilde{\boldsymbol{S}}_l\tilde{\boldsymbol{S}}_l^T\big) \tag{63}$$

$$\mathbf{S}_l = \mathbf{S}_l .* \big(\mathbf{\Omega}_{l-1}^T\mathbf{Y}\big)./\big(\mathbf{\Omega}_{l-1}^T\mathbf{\Omega}_{l-1}\mathbf{S}_l\big). \tag{64}$$

Then, the endmember and abundance obtained by the proposed algorithm are as follows:

$$\mathbf{X} = \mathbf{X}_1\mathbf{X}_2, \ldots, \mathbf{X}_L \tag{65}$$

$$\mathbf{S} = \mathbf{S}_L. \tag{66}$$

The abundance matrix $\mathbf{S} \in \mathbb{R}^{R\times IJ}$ is the abundance $\mathbf{S}_L$ of the last layer of multilayer decomposition, and $\mathbf{X} \in \mathbb{R}^{K\times R}$ is the product of all endmember matrices decomposed by L layers.

The nonnegative and sum-to-one constraints of abundance are directly imposed in the iteration. The iteration stopping criteria and initialization values are similar to Algorithm 1, and the proposed algorithm is summarized in Algorithm 2. The flowchart of the proposed algorithms is shown in Fig. 3.

IV. Experiments

In this section, the simulated data consisting of different endmembers and three real hyperspectral datasets with different sizes and complexities were used to validate the proposed algorithm. Some related state-of-the-art algorithms are used for comparison purposes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

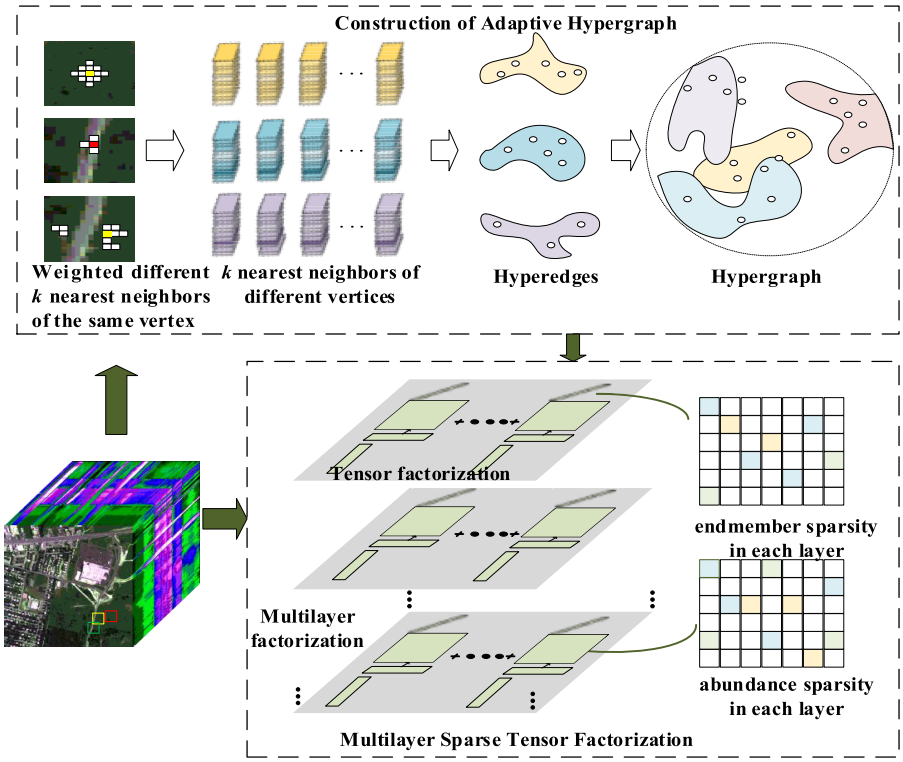


Fig. 3. Flowchart of the proposed algorithm.

A. Evaluation Indexes

Three performance indexes are adopted in order to evaluate the proposed method and comparison method: the SAD, the root-mean-square error (RMSE), and the signal to reconstruction error (SRE) [53]. The SAD is used to evaluate the difference between the estimated endmembers and true endmembers, defined as

$$\mathrm{SAD}_n = \arccos\left(\frac{\mathbf{c}_n^T \widehat{\mathbf{c}}_n}{\|\mathbf{c}_n\|\|\widehat{\mathbf{c}}_n\|}\right) \tag{67}$$

where $\mathbf{c}_n$ denotes the nth true endmember, and $\hat{\boldsymbol{c}}_n$ denotes the nth estimated endmember. The smaller the value of SAD, the higher the accuracy of the estimated endmember is. The RMSE is used to reveal the performance of the estimated abundance as

$$\mathrm{RMSE}_n = \sqrt{\frac{1}{N}\|\mathbf{E}_n - \widehat{\mathbf{E}_n}\|_2^2} \tag{68}$$

where $\mathbf{E}_n$ is the nth true abundance map, and $\hat{\mathbf{E}}_n$ is its corresponding estimated abundance map. N denotes the number of pixels in hyperspectral image. The smaller the value of RMSE, the closer the estimated abundance is to the true abundance map. The SRE is used to evaluate the unmixing performance between the hyperspectral image $\mathbf{Y}$ and the mixed image $\hat{\mathbf{Y}}$ from the corresponding estimated endmember and abundance map, which is defined as

$$\mathrm{SRE} = 10\log_{10}\frac{E\left[\|\mathbf{Y}\|_2^2\right]}{E\left[\|\mathbf{Y} - \widehat{\mathbf{Y}}\|_2^2\right]}. \tag{69}$$

B. Comparative Methods

To evaluate the effectiveness of the proposed methods, seven related and latest methods are selected for comparison.

1) The SUnSAL algorithm is a typical sparse unmixing algorithm [11]. It does not require endmember extraction, and the estimated abundance is directly obtained by using the spectral library. Only RMSE and SRE values can be calculated during performance comparison.
2) MV-NTF method is a tensor factorization with good unmixing performance [41], and the endmembers and abundances conform to the physical mechanism of mixed pixels.
3) Hypergraph regularized sparse nonnegative matrix factorization (HG$L_{1/2}$-NMF) constructs hypergraph structure to represent the similarity among spatial nearby pixels [28], and sparse constraint is imposed on abundances.
4) Multilayer nonnegative matrix factorization (MLNMF) [51] for spectral unmixing decomposes the hyperspectral image in several layers, and in each layer, the sparse constraint is applied on the endmembers and abundance maps.
5) Adaptive graph regularized multilayer nonnegative matrix factorization (AGMLNMF) [31] model constructed the graph on the probabilities between neighbors, and weights of the graph are assigned based on the relationships among neighbors.
6) Sparsity constrained deep nonnegative matrix factorization (SDNMF) [54] for unmixing adopted the concept of deep learning, the method used the fine-tuning stage

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

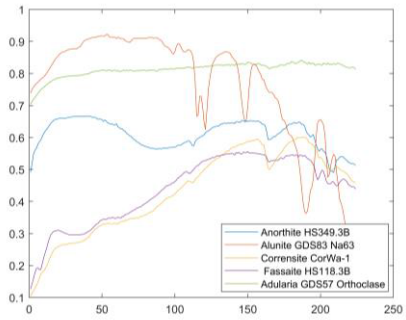


Fig. 4. Five endmembers of simulated data.

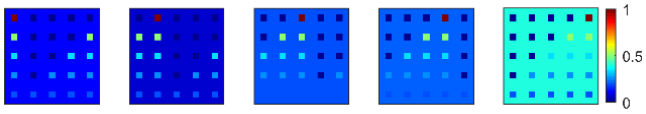


Fig. 5. Simulated data and abundance maps.

to reduce the total reconstruction error and enforce the sparse constraint on the abundance matrix.

7) Nonlinear low-rank nonnegative tensor decomposition algorithm (LRNTF) is proposed to solve the generalized bilinear model for unmixing [55]; since endmembers are used as prior information, only RMSE and SRE values can be calculated during performance comparison.

C. Parameter Setting

The parameters of the proposed unmixing algorithms include four regularization parameters and one Lagrange parameter. The k nearest neighbors in hypergraph construction and the number of layers L are also important parameters. The regularization parameters and Lagrange parameter are changed from 0, 1e-3, 1e-2, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9 to 1. The value of k is changed from 4 to 16. The range of layers is $\{2, 4, 6, 8, 10\}$. In the experiments, the parameters with the best unmixing results are selected. The parameters of the comparison algorithms are selected from the corresponding articles.

D. Simulated Data Experiments

The USGS spectral library was used to construct the ground-truth endmembers, with a spectral range of 0.4–2.5 μm. Five spectral features were randomly selected from USGS spectral library as endmembers to generate a linear simulation dataset. The dataset contains 75 $\times$ 75 pixels and each pixel contained 224 bands. Fig. 4 shows the spectra of five endmembers. Fig. 5 shows the simulated image and the real abundance map of five endmembers. These pixels are distributed in a fixed square area. The background pixels of the simulated image consist of a mixture of five identical endmembers, but their corresponding abundance coefficients are fixed at 0.1149, 0.0742, 0.2003, 0.2055, and 0.4051 [56]. The simulated hyperspectral images were generated according to the procedure described above, and the experiments are carried out under the condition of Gaussian white noise pollution. In this experiment, the signal-to-noise ratio (SNR) of simulated data is 25 dB.

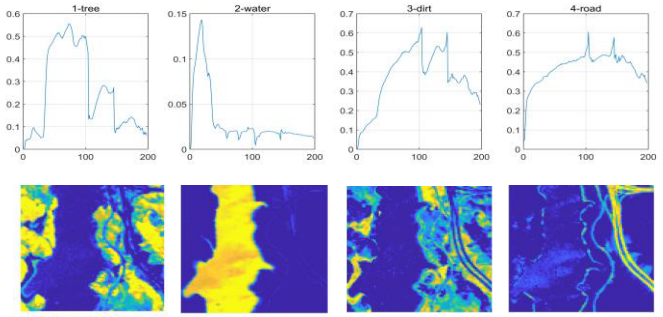


Fig. 6. Four ground-truth endmembers and abundance maps of Jasper data.

The true endmember and abundance of simulated data are known, and three performance metrics can be used to evaluate the unmixing performance of the proposed algorithms. The results with the highest accuracy in Table II are bolded. The results with the second-highest accuracy are underlined. As can be seen from Table II, for the RMSE results, the lower the value, the higher the accuracy of the endmembers is. The highest accuracy of these five endmembers is different but for the average RMSE results, the RMSE results are 0.1313 and 0.1129, respectively, with good accuracy. The RMSE result of the proposed AHGNTF algorithm is higher than that of the AHGMLSTF method, indicating that multilayer sparse decomposition is a positive impact on unmixing. Comparing the accuracy indicators of different algorithms, it can be seen that the unmixing performance of SDNMF is second only to the proposed algorithms, and SDNMF algorithm and the proposed algorithms are used for multilayer decomposition ideas, which can speculate that this concept is effective for unmixing. The accuracy of the two proposed algorithms is higher than that of the original tensor factorization algorithm MV-NTF, indicating that the proposed regularization items are effective. Compared with other algorithms, the proposed AHGNTF and AHGMLSTF methods have better accuracy.

E. Real Data Experiments

1) Jasper Ridge Data: Jasper Ridge data are popular for hyperspectral unmixing, with relatively simple land cover. It was collected over Jasper Ridge Natural Reserve in CA, USA. The data consist of 512 $\times$ 614 pixels, and it has 224 bands. Its spectral resolution is 9.46 nm, and the spectral range is from 380 to 2500 nm. For experiments, an area with 100 $\times$ 100 pixels is cropped. To avoid atmospheric effects, 26 bands were removed and 198 bands were retained. There are four endmembers: road, dirt, water, and tree. Using the data from [57], true abundance and endmember for quantitative analysis are shown in Fig. 6.

The results of Jasper data are calculated using the comparative algorithms and the proposed algorithms, and the results obtained are shown in Table III. SUnSAL algorithm directly obtains the abundance results without endmember extraction, so only the RMSE results can be calculated. As can be seen from Table II that the proposed algorithms AHGNTF and AHGMLSTF have the best unmixing performance for water, adaptive graph method AGMLNMF and proposed adaptive hypergraph algorithms perform well in the unmixing of roads, and the multilayer decomposition MLNMF, SDNMF, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

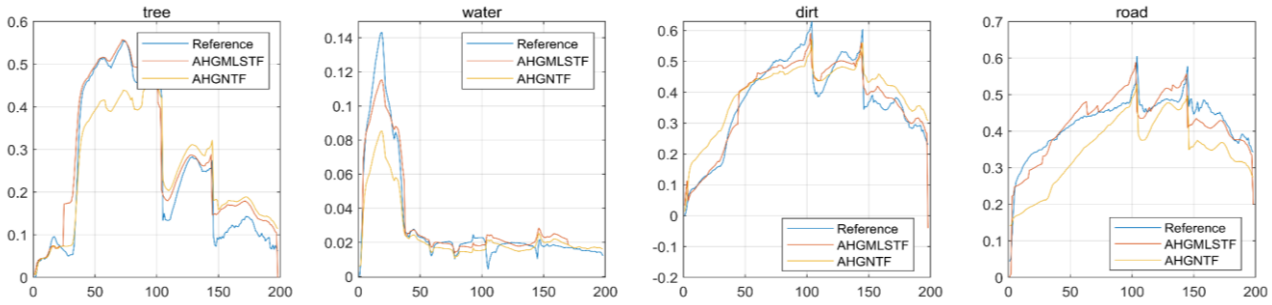


Fig. 7. Estimated endmembers by the proposed AHGMLSTF and AHGNTF methods and the reference endmembers on Jasper data.

TABLE II
RMSE, SAD, AND SRE RESULTS OF COMPARISON ALGORITHMS ON SIMULATED DATA

	Method	LRNTF	SUnSAL	MV-NTF	HG $L_{1/2}$ -NMF	MLNMF	AGMLNMF	SDNMF	AHGNTF	AHGMLSTF
RMSE	Anorthite HS349.3B	0.2392	0.407	0.1777	**0.0435**	0.3396	0.0066	0.1985	0.1081	0.1699
	Alunite GDS83 Na63	0.1291	**0.0502**	0.2166	0.0940	0.3720	0.1114	0.1173	0.1195	0.1467
	Corrensite CorWa-1	0.1310	0.3192	0.1555	0.2277	0.3795	0.0440	0.0587	0.1083	**0.0385**
	Fassaite HS118.3B	0.1423	0.2674	0.1746	0.1162	**0.0312**	0.3818	0.2295	0.2223	0.1201
	Adularia GDS57 Orthoclase	0.0810	0.4455	**0.0574**	0.3070	0.0730	0.2826	0.0744	0.0984	0.0892
	Average	0.1445	0.2979	0.1564	0.1577	0.2391	0.1653	0.1357	0.1313	**0.1129**
SAD		-	-	0.173	0.3251	0.201	0.2121	0.1593	0.1355	**0.052**
SRE		27.5092	25.5923	21.5231	29.6342	26.2947	23.642	30.4913	31.2601	**38.3235**

TABLE III
RMSE, SAD, AND SRE RESULTS OF COMPARISON ALGORITHMS ON JASPER DATA

	Method	LRNTF	SUnSAL	MV-NTF	HG $L_{1/2}$ -NMF	MLNMF	AGMLNMF	SDNMF	AHGNTF	AHGMLSTF
Tree	RMSE	0.1583	0.1697	0.3867	0.7302	0.5736	0.656	**0.1383**	0.3104	0.1521
	SAD	/	/	0.2324	0.3424	0.3029	0.0907	0.3528	0.0623	**0.0079**
Water	RMSE	0.1329	0.4689	0.2976	0.2506	0.63	0.1578	0.3482	0.0755	**0.0678**
	SAD	/	/	0.3435	0.6086	0.3817	0.4424	**0.139**	0.2607	0.2034
Dirt	RMSE	0.2219	0.568	0.3957	0.4825	0.2834	0.5934	0.8957	0.2882	**0.1377**
	SAD	/	/	0.3354	0.2296	0.168	0.2542	0.5885	**0.0853**	0.2464
Road	RMSE	0.2948	0.3495	0.5704	0.6875	0.2472	**0.0496**	0.1161	0.0734	0.1507
	SAD	/	/	0.3382	0.5202	0.4248	0.1406	**0.0316**	0.288	0.177
Average	RMSE	0.2019	0.389	0.4126	0.5377	0.4335	0.3642	0.3746	0.1869	**0.1271**
	SAD	/	/	0.3124	0.4252	0.3193	0.232	0.278	0.1741	**0.1587**
SRE		33.9807	/	30.7522	28.424	15.7524	24.5832	33.6243	34.735	**42.3591**

AHGMLSTF algorithms all perform well in the unmixing of soil and tree. Comparing the average value of RMSE and SAD, the proposed algorithms have better accuracy, and the AHGMLSTF method is better than the AHGNTF method for the unmixing of Jasper data. In general, adaptive hypergraph and multilayer decomposition sparse term have a positive effect on unmixing, and the proposed algorithms have better unmixing performance. The endmembers obtained by comparison methods and reference endmembers are shown in Fig. 7. The estimated abundance maps and ground-truth abundance maps are shown in Fig. 8.

2) Urban Data: Urban data are one of the most commonly used hyperspectral data in hyperspectral unmixing and have a complex land cover distribution. It was acquired by the HYDICE sensor and contains 307 × 307 pixels. In this image, the spectral range is from 400 to 2500 nm, with a spectral resolution of 10 nm. After channels 1–4, 76, 87, 101–111, 136–153, and 198–210 were removed (due to dense water vapor and atmospheric effects), 162 channels remained. There are four endmembers in the scene: asphalt, tree, grass, and roof. The endmembers and corresponding abundance are shown in Fig. 9.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RMSE, SAD, AND SRE RESULTS OF COMPARISON ALGORITHMS ON URBAN DATA

	Method	LRNTF	SUnSAL	MV-NTF	HG $L_{1/2}$ -NMF	MLNMF	AGMLNMF	SDNMF	AHGNTF	AHGMLSTF
Road	RMSE	0.4799	0.3051	0.6982	0.1992	0.3207	0.5824	0.1950	**0.1175**	0.5187
	SAD	/	/	0.4024	0.2917	0.7515	0.4447	0.5060	**0.1428**	0.1584
Grass	RMSE	0.3492	0.7067	0.4448	0.7906	0.7201	0.4796	0.9161	0.5090	**0.0011**
	SAD	/	/	0.9364	0.3643	0.3326	0.9679	0.5553	**0.0518**	0.6484
Tree	RMSE	0.3637	0.8388	**0.0254**	0.5154	0.9976	0.1956	0.5003	0.2878	0.4007
	SAD	/	/	0.7266	0.8689	0.4614	**0.0665**	0.9322	0.6700	0.1379
Roof	RMSE	0.4272	0.5024	0.7114	0.6776	0.4626	0.3093	**0.2990**	0.6321	0.4178
	SAD	/	/	0.4609	0.7149	0.5504	0.4240	**0.1132**	0.9183	0.2589
Average	RMSE	0.4050	0.5882	0.4699	0.5457	0.6252	0.3917	0.4776	0.3866	**0.3346**
	SAD	/	/	0.6316	0.5599	0.5240	0.4758	0.5267	0.4457	**0.3009**
SRE		34.6920	/	24.2900	28.3165	13.8554	29.4921	26.8431	41.3955	**45.0064**

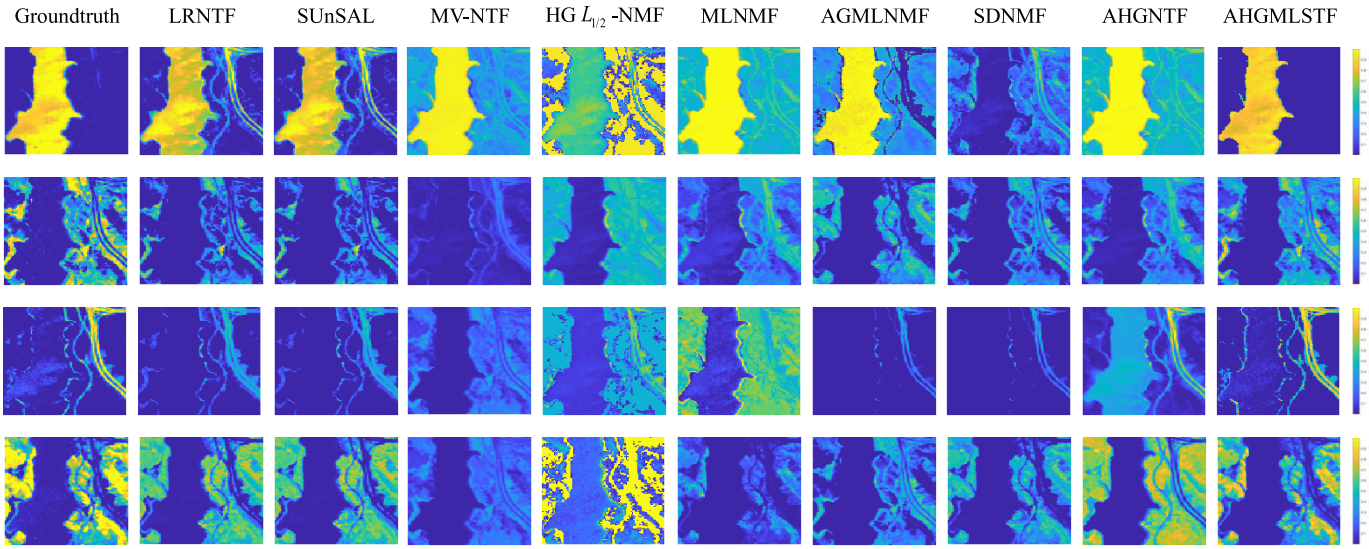


Fig. 8. Estimated abundance maps by comparison algorithms and the ground-truth abundance maps on Jasper data.

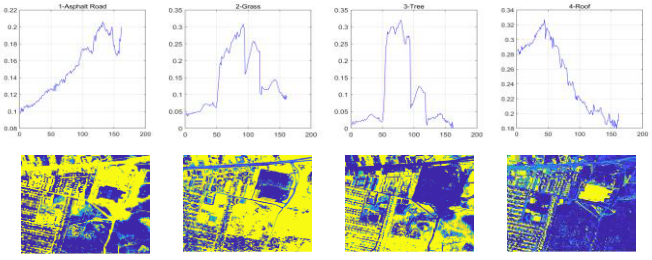


Fig. 9. Four ground-truth endmembers and abundance maps of Urban data.

Overall, the average RMSE, SAD, and SRE results of the proposed AHGMLSTF method are 0.3346, 0.3009, and 45.004, respectively, and the results of AHGNTF algorithms are 0.3866, 0.4457, and 41.3955, respectively. In Table IV, it can be observed that the accuracy of both proposed algorithms is higher than the results of the comparison algorithms. For single endmembers, tensor factorization algorithms MV-NTF and the proposed algorithms have better performance in grass and tree. The algorithm based on the adaptive graph AGML-NMF and proposed adaptive hypergraph methods AHGNTF and AHGMLSTF can well perform on the roof. In general, the two regularization items of the proposed method have effective effects on unmixing. The estimated endmembers and abundance maps are shown in Figs. 10 and 11, respectively.

3) Chang'E-3 VNIS Data: This hyperspectral data was acquired by Visible and Near-Infrared Imaging Spectrometer (VNIS) on the lunar rover, located in the northwest of northern Imbrium over the lunar surface. The image contains 256 × 256 pixels with 100 bands. The spectral range is from 450 to 950 nm, and its spectral resolution is 5 nm. The level 2B Chang'E-3 VNIS data are available at https://moon.bao.ac.cn/web/zhmanager/home in PDS format, and VNIS data need to be preprocessed, including bad lines, bad points removal, flat field correction, and reflectivity conversion [58].

Since there is no ground truth, only SRE can be calculated to evaluate algorithm performance. It can be seen from Table V that the proposed algorithms have the highest SRE value on CE-3 VNIS data, the SRE of AGMLNMF is second only to the proposed algorithms, and the unmixing performance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

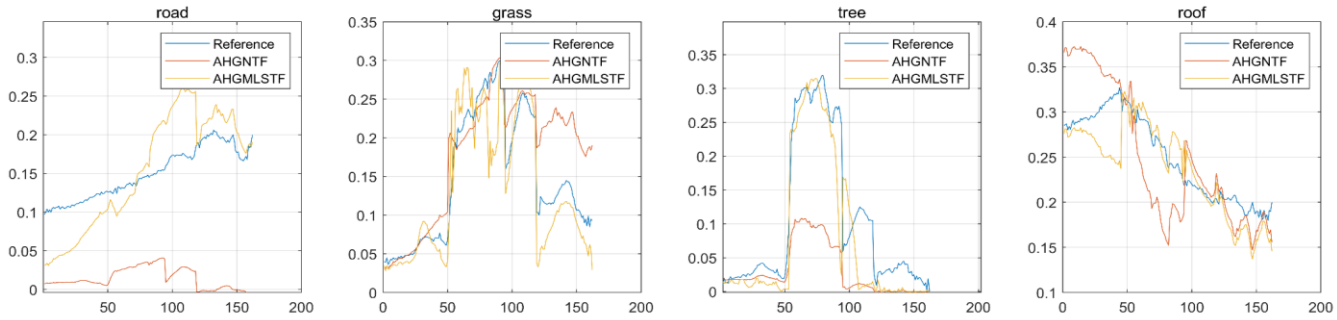


Fig. 10. Estimated endmembers by the proposed AHGMLSTF and AHGNTF methods and the reference endmembers on Urban data.

TABLE V
SRE RESULTS OF COMPARISON ALGORITHMS ON CE-3 VNIS DATA

	MV-NTF	LRNTF	HG $L_{1/2}$-NMF	MLNMF	AGMLNMF	SDNMF	AHGNTF	AHGMLSTF
SRE	17.4561	20.382	22.57	14.4514	24.0739	22.7461	24.4881	**27.1565**

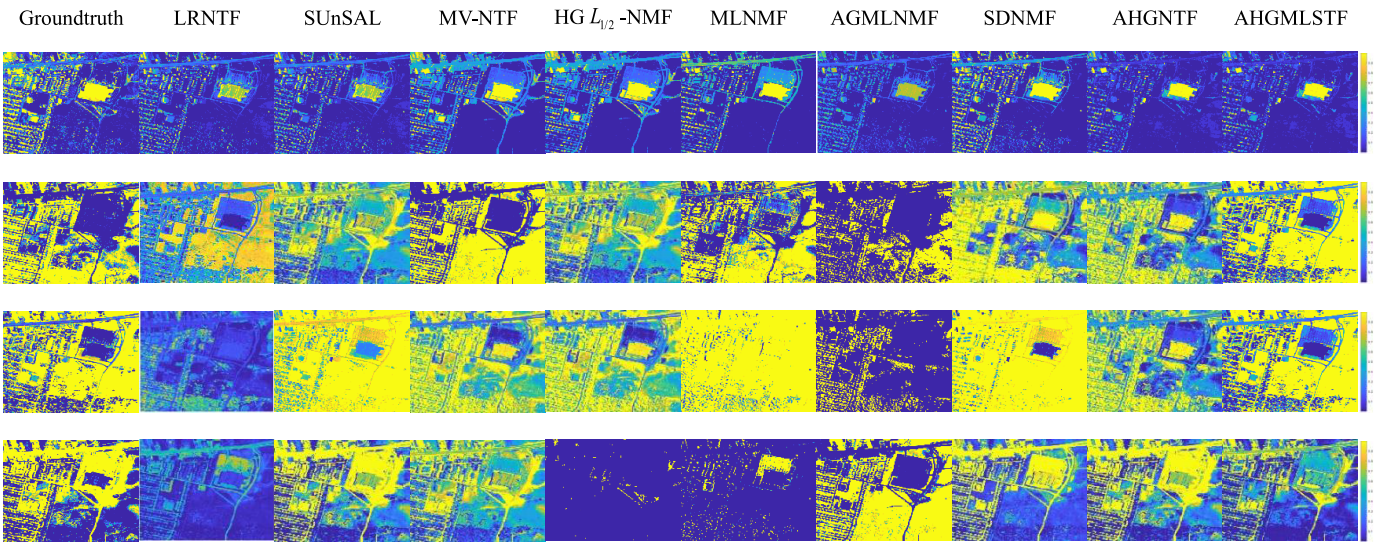


Fig. 11. Estimated abundance maps by comparison algorithms and the ground-truth abundance maps on Urban data.

MLNMF with only the multilayer decomposition concept is the lowest. The endmembers and abundance maps obtained are shown in Figs. 12 and 13, respectively.

The four sets of experiments were carried out to verify the performance of the proposed algorithms. For simulated data mixed with different endmembers, the proposed algorithm has better unmixing performance compared to the comparative algorithms. For two hyperspectral images with simple ground distribution and complex ground distribution, it can be seen that the proposed algorithms using spectral–spatial information offer high unmixing accuracy.

V. DISCUSSION

The proposed method constructs an adaptive hypergraph to represent the joint spectral–spatial information and uses the multilayer decomposition method for unmixing; the sparsity constraint is imposed on abundance and endmember in each layer. The experiments in this section mainly discuss the effectiveness of the above contributions.

A. Parameter Analysis

The regularization parameters in the proposed method control the influence of the adaptive hypergraph regularization term and sparse term for unmixing; it is necessary to analyze the sensitivity of the proposed method to the regularization parameters and find out the appropriate parameter range for practical application and subsequent analysis experiments. There are two parameters δ and γ to control the adaptive hypergraph regularization term and two parameters λ and β to control sparsity; since $\lambda = 2\beta$, only λ is set to change, and β will change accordingly.

The simulated data are constructed according to the method mentioned above. Parameters δ, λ, and γ vary in the range of $\{0, 0.001, 0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$. Fig. 14(a) shows the RMSE results. It can be observed that the value of RMSE is the largest when the regularized parameters are 0, indicating that the regularization parameters are effective. When $\gamma = 0$, the RMSE results are bigger than 0.4. When $\lambda = 0$, the RMSE results are larger than 0.3, revealing that the adaptive hypergraph term controlled by γ has a more positive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

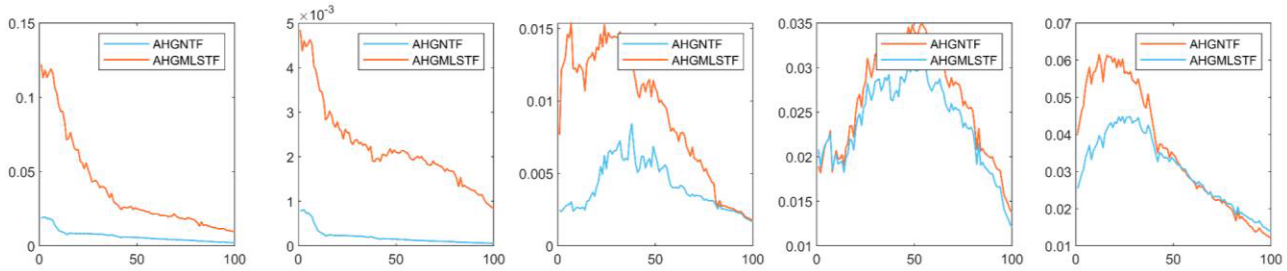


Fig. 12. Estimated endmembers by the proposed AHGMLSTF and AHGNTF methods on CE-3 VNIS data.

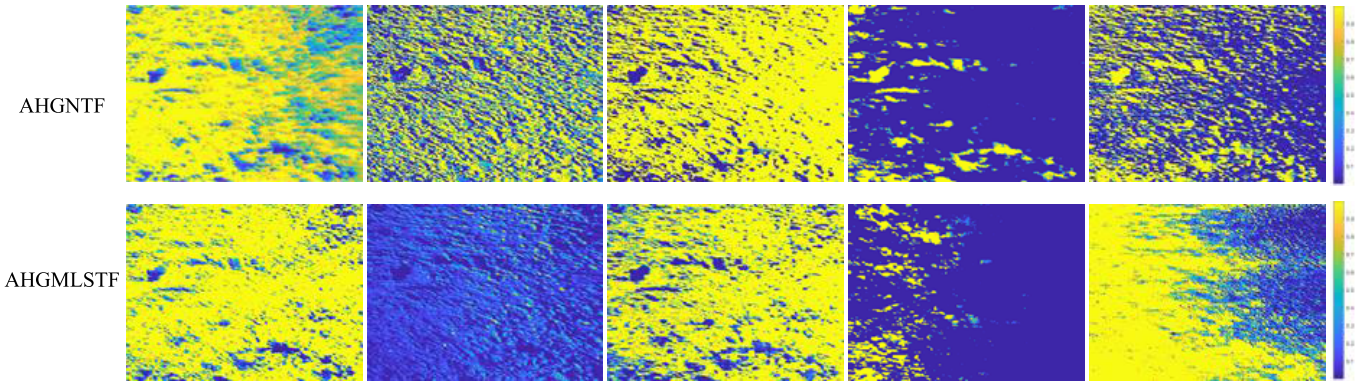


Fig. 13. Estimated abundance maps by the proposed algorithms on CE-3 VNIS data.

TABLE VI
RMSE, SAD, AND SRE RESULTS OF COMPARISON ALGORITHMS ON DIFFERENT DISTANCE

Algorithms	AHGNTF (Euclidean distance)	AHGMLSTF (Euclidean distance)	AHGNTF(SAD)	AHGMLSTF(SAD)
RMSE	0.1587	0.1464	0.1313	0.1129
SAD	0.1479	0.1341	0.1355	0.052
SRE	29.9093	32.4806	31.2601	38.3235

effect on unmixing. As the regularization parameters λ and γ vary from 0 to 1, the RMSE results of the AHGMLSTF algorithm decrease first and then increase. The proposed AHGMLSTF method performs well in the range of $\lambda = [0.3, 0.4]$ and $\gamma = [0.7, 0.8]$. With $\lambda = 0.3$, δ and γ vary in the range of $\{0, 0.001, 0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$. Both δ and γ are adaptive hypergraph regularized parameters. As shown in Fig. 14(b), the RMSE results decrease as the parameters δ and γ vary from 0 to 1; the proposed AHGMLSTF method performs well in the range of $\delta = [0.5, 1]$ and $\gamma = [0.6, 1]$. When the two parameters are 0, the RMSE result is the largest, indicating the validity of the regularization terms controlled by the two parameters.

B. Effectiveness Analysis of Adaptive Hypergraph Construction

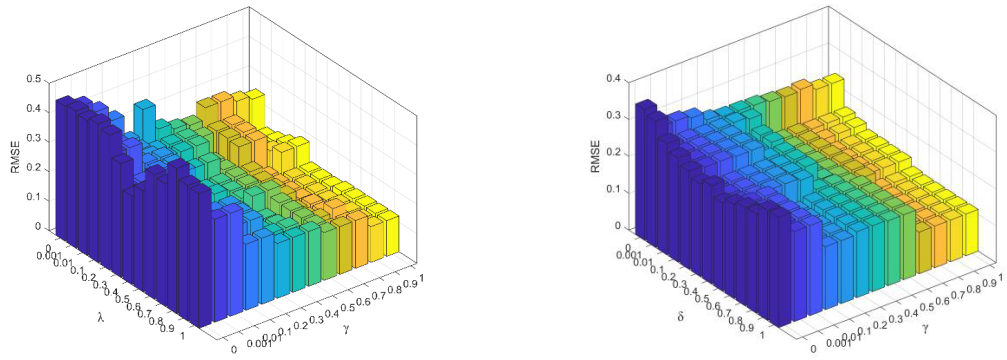
The construction method of the hypergraph in the proposed algorithms has been improved. To verify the effectiveness of the SAD distance for unmixing, the hypergraph tensor decomposition algorithm using Euclidean distance is compared with the hypergraph tensor decomposition algorithm constructed by SAD distance in this article. In Table VI, it can be seen that the tensor decomposition algorithm using the SAD distance AHGNTF and AHGMLSTF algorithms have better performance in both endmember extraction and abundance estimation.

In the proposed AHGMLSTF method, a set of hyperedges with k nearest neighbors are constructed to generate an adaptive hypergraph. In order to evaluate the effectiveness of the adaptive hypergraph, k nearest neighbors are fixed and the resultant hypergraph is compared with those constructed from a set of different k nearest neighbors. Fig. 15 shows three assessment index results of adaptive and fixed values of k. As k changes, the accuracy of both algorithms increases first and then decreases, and the proposed AHGMLSTF method performs best when $k = 12$. For these three indicators, the RMSE and SAD results of the algorithm with fixed k are higher than those of the adaptive k algorithm, and the SRE results are lower than those of the adaptive algorithm, indicating that the proposed adaptive k algorithm has good unmixing performance.

C. Effectiveness Analysis of Multilayer Factorization

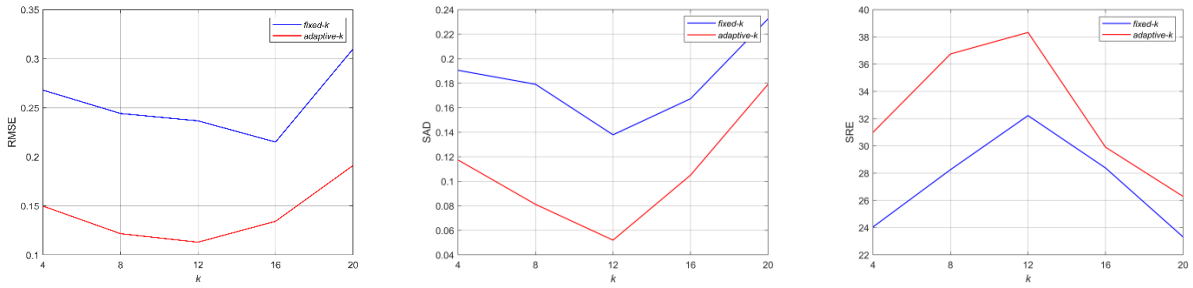
In the proposed AHGMLSTF method, the multilayer concept is introduced for hyperspectral unmixing. A set

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.



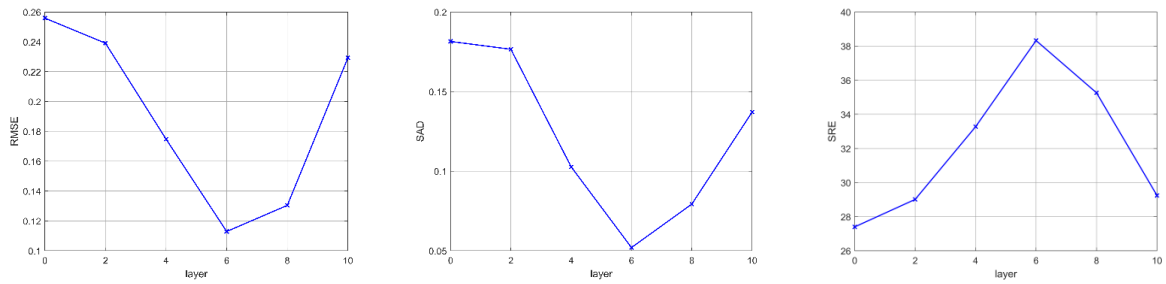
(a) (b)

Fig. 14. RMSE results of AHGMLSTF algorithm when δ, λ, and γ vary from 0.001 to 1: (a) RMSE results under different λ and γ and (b) RMSE results under different δ and γ.



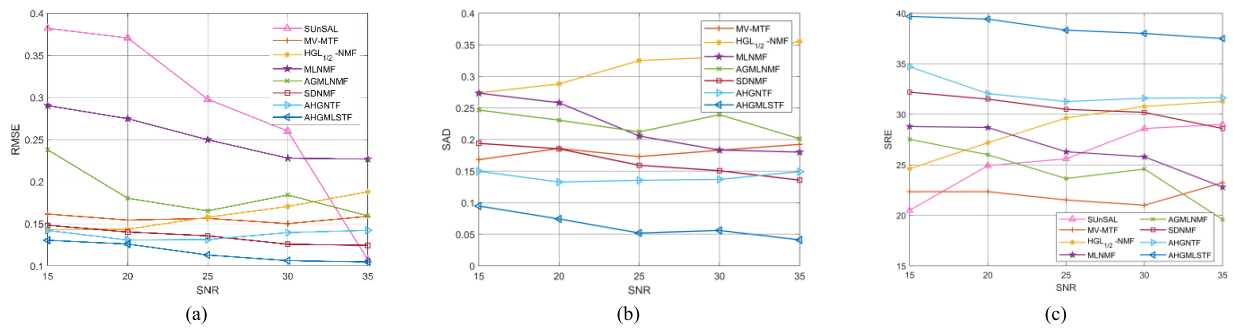
(a) (b) (c)

Fig. 15. Performance of adaptive hypergraph-based AHGMLSTF algorithm and fixed k algorithm: (a) RMSE; (b) SAD; and (c) SRE.



(a) (b) (c)

Fig. 16. Performance of the proposed AHGMLSTF algorithm under different layer numbers: (a) RMSE; (b) SAD; and (c) SRE.

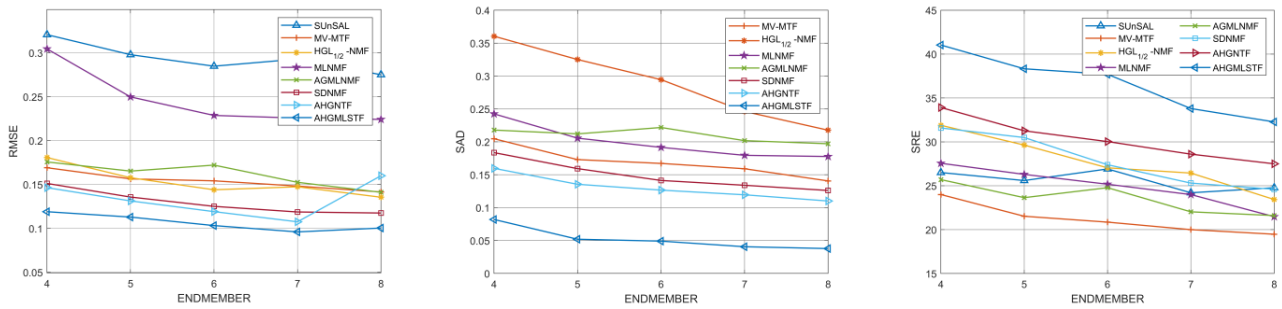


(a) (b) (c)

Fig. 17. Performance comparison of different algorithms under different levels of SNR: (a) RMSE; (b) SAD; and (c) SRE.

of experiments are carried out to verify the effectiveness of multilayer decomposition in the proposed algorithm by changing the number of layers of factorization. As can be seen from Fig. 16, when the number of layers is 0, that is, the multilayer decomposition is not performed, and the accuracy is lower than that of multilayer decomposition. RMSE, SAD, and SRE results have the same trend; when the number of layers increases, the accuracy of unmixing increases first and then decreases, and the proposed AHGMLSTF method performs best when the number of layers is 6. In general,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 18. Performance comparison of different algorithms under different endmember numbers: (a) RMSE; (b) SAD; and (c) SRE.

TABLE VII
RUNTIME OF COMPARISON ALGORITHMS ON DIFFERENT DATA

	Dataset	LRNTF	SUnSAL	MV-NTF	HG $L_{1/2}$-NMF	MLNMF	AGMLNMF	SDNMF	AHGNTF	AHGMLSTF
	Simulated data	34.31	0.11	45.23	25.8332	18.53	52.35	41.23	59.13	94.42
Runtime	Jasper	521.42	0.18	1648.61	416.24	348.94	452.18	729.24	932.24	1729.23
(second)	Urban	832.63	1.28	1729.42	341.44	525.42	490.73	881.68	1229.22	2945.53
	CE-3	379.43	0.81	136.15	179.42	292.45	214.59	329.27	1019.53	2432.31

multilayer factorization is effective for the proposed tensor factorization unmixing algorithm.

D. Robustness to Noise

The objective of this group of experiments is to verify the noise robustness of the proposed algorithms. The method of constructing the simulated data is as mentioned above, with the SNR being 15, 20, 25, 30, and 35 dB. As shown in Fig. 17, the proposed AHGNTF and AHGMLSTF algorithms have the best accuracy in abundance and endmember estimated, and the SRE results of the proposed algorithms are also the highest. When the SNR varies from 15 to 35, tensor factorization-based methods MV-NTF, AHGNTF, and AHGMLSTF algorithms outperform, indicating that the tensor has maintained the structure of hyperspectral data. The proposed methods are more robust to different levels of noise than the comparison algorithms since the adaptive hypergraph applied to the multilayer factorization can capture the joint spectral–spatial information.

E. Influence of Number of Endmembers

The objective of this group of experiments is to build simulation data with different numbers of endmembers to verify the effectiveness of the proposed algorithms. In the experiments, the endmember number varies from 4 to 8. As shown in Fig. 18, regardless of the number of endmembers, the accuracy of the proposed algorithms is higher than the comparative methods. SDNMF also performs well in different numbers of endmembers of simulated data, showing that the multilayer decomposition can represent the characteristics of mixed pixels. The adaptive hypergraph can extract the local spatial–spectral information, making the proposed algorithms perform best in simulated data with different numbers of endmembers.

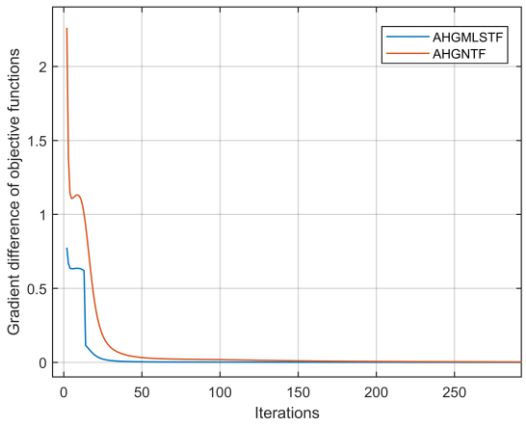


Fig. 19. Empirical convergence analysis of the proposed algorithms on simulated data.

F. Empirical Convergence Analysis and Runtime

In order to analyze the empirical convergence of the proposed algorithms, Fig. 19 shows the convergence curve of the proposed algorithms in simulated data. It can be seen that the gradient difference of the objective function between the current and the previous iterations drops quickly before iteration 100, almost unchanged and close to 0 after iteration 200.

From the running time of each comparison algorithm on different data in Table VII, it can be seen that the sparse unmixing algorithm SUnSAL takes the shortest time, the nonnegative matrix-based factorization methods take more time, and the tensor-based decomposition algorithms take the longest time due to more variables and the complex update rules.

VI. CONCLUSION

In this article, a novel AHGMLSTF algorithm is proposed for hyperspectral unmixing. In the proposed method, the hypergraph uses SAD instead of Euclidean distance to construct hyperedges, and the hypergraph is constructed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

adaptively by varying the k nearest neighbors of hyperedges, which can better represent the joint spatial and spectral information. In order to explore the hierarchical features of hyperspectral images, the images are decomposed into multilayers, and a sparse constraint is imposed on each layer to make the unmixing results more consistent with the physical mechanism of mixed spectral pixels. With these constraints, the proposed method represents not only the local neighborhood similarity but also the heterogeneity of adjacent edges of the hyperspectral images. Experiments on simulated data and real hyperspectral data demonstrate that the adaptive hypergraph regularization term and multilayer sparse factorization of the proposed method are effective for unmixing.

ACKNOWLEDGMENT

The authors thank the Chang'E-3 payload team for mission operations and China National Space Administration (Beijing, China) for providing the Chang'E-3 data. The Chang'E-3 data used in this work are processed and produced by Ground Research and Application System (GRAS) of China's Lunar and Planetary Exploration Program.

REFERENCES

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[3] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[4] W.-K. Ma et al., "A signal processing perspective on hyperspectral unmixing: Insights from remote sensing," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 67–81, Jan. 2013.

[5] A. Plaza, P. Martínez, R. Pérez, and J. Plaza, "A quantitative and comparative analysis of endmember extraction algorithms from hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 650–663, Mar. 2004.

[6] J. B. Adams, J. B. Adams, J. B. Adams, and A. R. Gillespie, *Remote Sensing of Landscapes With Spectral Images: A Physical Modeling Approach*. Cambridge, U.K.: Cambridge Univ. Press, 2006.

[7] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," in *Proc. SPIE*, vol. 3753, 1999, pp. 266–275.

[8] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[9] J. Li and J. M. Bioucas-Dias, "Minimum volume simplex analysis: A fast algorithm to unmix hyperspectral data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2008, pp. 1–4.

[10] C.-I. Chang and D. Heinz, "Constrained subpixel target detection for remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1144–1159, May 2000.

[11] J. M. Bioucas-Dias, "A variable splitting augmented Lagrangian approach to linear spectral unmixing," in *Proc. 1st Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, 2009, pp. 1–4.

[12] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2013.

[13] R. Wang, H.-C. Li, W. Liao, and A. Pizurica, "Double reweighted sparse regression for hyperspectral unmixing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 6986–6989.

[14] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2011.

[15] R. Feng, Y. Zhong, and L. Zhang, "Adaptive spatial regularization sparse unmixing strategy based on joint MAP for hyperspectral remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 12, pp. 5791–5805, Dec. 2016.

[16] Y. Zhong, R. Feng, and L. Zhang, "Non-local sparse unmixing for hyperspectral remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1889–1909, Jun. 2014.

[17] F. Li, S. Zhang, C. Deng, B. Liang, J. Cao, and S. Wang, "Robust double spatial regularization sparse hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 12569–12582, 2021.

[18] J. Li, Y. Li, R. Song, S. Mei, and Q. Du, "Local spectral similarity preserving regularized robust sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7756–7769, Oct. 2019.

[19] H. Su, W. Yao, Z. Wu, P. Zheng, and Q. Du, "Kernel low-rank representation with elastic net for China coastal wetland land cover classification using GF-5 hyperspectral imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 171, pp. 238–252, Jan. 2021.

[20] G. Zhang et al., "Spectral variability augmented sparse unmixing of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527413.

[21] H. Su, C. Jia, P. Zheng, and Q. Du, "Superpixel-based weighted collaborative sparse regression and reweighted low-rank representation for hyperspectral image unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 393–408, 2022.

[22] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.

[23] P.-A. Thouvenin, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral unmixing with spectral variability using a perturbed linear mixing model," *IEEE Trans. Signal Process.*, vol. 64, no. 2, pp. 525–538, Jan. 2016.

[24] L. Drumetz, M.-A. Veganzones, S. Henrot, R. Phlypo, J. Chanussot, and C. Jutten, "Blind hyperspectral unmixing using an extended linear mixing model to address spectral variability," *IEEE Trans. Image Process.*, vol. 25, no. 8, pp. 3890–3905, Aug. 2016.

[25] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2012.

[26] S. Yang, X. Zhang, Y. Yao, S. Cheng, and L. Jiao, "Geometric nonnegative matrix factorization (GNMF) for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2696–2703, Jun. 2015.

[27] L. Tong, J. Zhou, Y. Qian, and Y. Gao, "Multiple graph regularized NMF for hyperspectral unmixing," in *Proc. 7th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2015, pp. 1–4.

[28] W. Wang, Y. Qian, and Y. Y. Tang, "Hypergraph-regularized sparse NMF for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 681–694, Feb. 2016.

[29] X. Li, X. Zhang, Y. Yuan, and Y. Dong, "Adaptive relationship preserving sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504516.

[30] M. Zhao, T. Gao, J. Chen, and W. Chen, "Hyperspectral unmixing via nonnegative matrix factorization with handcrafted and learned priors," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] L. Tong, J. Zhou, B. Qian, J. Yu, and C. Xiao, "Adaptive graph regularized multilayer nonnegative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 434–447, 2020.

[32] J. Peng, Y. Zhou, W. Sun, Q. Du, and L. Xia, "Self-paced nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1501–1515, Feb. 2020.

[33] Y. Bu et al., "Hyperspectral and multispectral image fusion via graph Laplacian-guided coupled tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 648–662, Jan. 2021.

[34] J. Xue, Y. Zhao, W. Liao, and J. C.-W. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.

[35] H. Zeng, X. Xie, H. Cui, H. Yin, and J. Ning, "Hyperspectral image restoration via global L_{1-2} spatial–spectral total variation regularized local low-rank tensor recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3309–3325, Jul. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.

[36] I. V. Oseledets, "Tensor-train decomposition," *SIAM J. Sci. Comput.*, vol. 33, no. 5, pp. 2295–2317, Jan. 2011.

[37] N. Renard, S. Bourennane, and J. Blanc-Talon, "Denoising and dimensionality reduction using multilinear tools for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 138–142, Apr. 2008.

[38] A. Huck and M. Guillaume, "Estimation of the hyperspectral tucker ranks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Apr. 2009, pp. 1281–1284.

[39] Q. Zhang, H. Wang, R. Plemmons, and V. P. Pauca, "Spectral unmixing using nonnegative tensor factorization," in *Proc. 45th Annu. Southeast Regional Conf.*, Mar. 2007, pp. 531–532.

[40] M. A. Veganzones, J. E. Cohen, R. C. Farias, J. Chanussot, and P. Comon, "Nonnegative tensor CP decomposition of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2577–2588, May 2016.

[41] Y. Qian, F. Xiong, S. Zeng, J. Zhou, and Y. Y. Tang, "Matrix-vector nonnegative tensor factorization for blind unmixing of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1776–1792, Mar. 2017.

[42] F. Xiong, Y. Qian, J. Zhou, and Y. Y. Tang, "Hyperspectral unmixing via total variation regularized nonnegative tensor factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2341–2357, Apr. 2019.

[43] P. Zheng, H. Su, and Q. Du, "Sparse and low-rank constrained tensor factorization for hyperspectral image unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1754–1767, 2021.

[44] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Apr. 2019.

[45] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509612.

[46] S. Mei, X. Chen, Y. Zhang, J. Li, and A. Plaza, "Accelerating convolutional neural network-based hyperspectral image classification by step activation quantization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502012.

[47] J. Xue, Y. Zhao, S. Huang, W. Liao, and S. G. Kong, "Multilayer sparsity-based tensor decomposition for low-rank tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 99, pp. 1–15, 2021.

[48] J. Xue, Y. Zhao, Y. Bu, J. C.-W. Chan, and S. G. Kong, "When Laplacian scale mixture meets three-layer transform: A parametric tensor sparsity for tensor completion," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 13887–13901, Dec. 2022, doi: 10.1109/TCYB.2021.3140148.

[49] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, Jan. 2011.

[50] C.-I. Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.

[51] R. Rajabi and H. Ghassemian, "Spectral unmixing of hyperspectral imagery using multilayer NMF," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 38–42, Jan. 2015.

[52] A. Cichocki, R. Zdunek, S. Choi, R. Plemmons, and S.-I. Amari, "Novel multi-layer non-negative tensor factorization with sparsity constraints," *Adaptive and Natural Computing Algorithms*, vol. 4432. Warsaw, Poland: Springer, 2007, pp. 271–280.

[53] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Aug. 2002.

[54] X.-R. Feng, H.-C. Li, J. Li, Q. Du, A. Plaza, and W. J. Emery, "Hyperspectral unmixing using sparsity-constrained deep nonnegative matrix factorization with total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6245–6257, Oct. 2018.

[55] L. Gao, Z. Wang, L. Zhuang, H. Yu, B. Zhang, and J. Chanussot, "Using low-rank representation of abundance maps and nonnegative tensor factorization for hyperspectral nonlinear unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504017.

[56] M.-D. Iordache, "A sparse regression approach to hyperspectral unmixing," Ph.D. dissertation Inst. Super. TÉCNICO, Dep. Electr. Comput. Eng., Lisbon, Portugal, 2011.

[57] F. Zhu, "Hyperspectral unmixing: Ground truth labeling, datasets, benchmark performances and survey," 2017, *arXiv:1708.05125*.

[58] B. Liu et al., "Data processing and preliminary results of the Chang'e-3 VIS/NIR imaging spectrometer in-situ analysis," *Res. Astron. Astrophys.*, vol. 14, no. 12, pp. 1578–1594, Dec. 2014.



Pan Zheng received the B.E. degree in geodesy and geomatics engineering and the M.E. degree in photogrammetry and remote sensing from the School of Earth Sciences and Engineering, Hohai University, Nanjing, China, in 2018 and 2021, respectively, where she is currently pursuing the Ph.D. degree in surveying and mapping.

Her research interests include hyperspectral remote sensing imagery, collaborative representation, and machine learning in hyperspectral unmixing.



Hongjun Su (Senior Member, IEEE) received the Ph.D. degree in cartography and geography information system from the Key Laboratory of Virtual Geographic Environment (Ministry of Education), Nanjing Normal University, Nanjing, China, in 2011.

He is currently a Full Professor with the School of Earth Sciences and Engineering, Hohai University, Nanjing. His main research interests include hyperspectral remote sensing dimensionality reduction, classification, and spectral unmixing.

Dr. Su received the 2016 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. He is an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Hongliang Lu received the B.E. degree from Chuzhou University, Chuzhou, China, in 2015, and the M.E. degree from the School of Geomatics, Anhui University of Science and Technology, Huainan, China, in 2020. He is currently pursuing the Ph.D. degree in surveying and mapping with the School of Earth Sciences and Engineering, Hohai University, Nanjing, China.

His research interests include ensemble learning, few shot learning, and collaborative representation in hyperspectral remote sensing imagery classification.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore, MD, USA, in 2000.

She is currently the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of SPIE—International Society for Optics and Photonics. She was a Co-Chair of the Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society from 2009 to 2013, and the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She has served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and the IEEE SIGNAL PROCESSING LETTERS. From 2016 to 2020, she was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She is currently a member of IEEE Periodicals Review and Advisory Committee.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:12:50 UTC from IEEE Xplore. Restrictions apply.